

AI/BigData 인텐시브 과정

강사장철원

Section 0

코스소개

DAY1

DAY2

DAY3

DAY4

DAY5

데이터 분석을 위한 기초 수학

데이터 집계 및 시각화

DAY6

DAY7

DAY8

DAY9

DAY10

미니
프로젝트

확률분포, AB테스트

DAY11

DAY12

DAY13

DAY14

DAY15

머신러닝

DAY16

DAY17

DAY18

DAY19

DAY20

딥러닝

파이널 프로젝트

□ matplotlib

□ seaborn

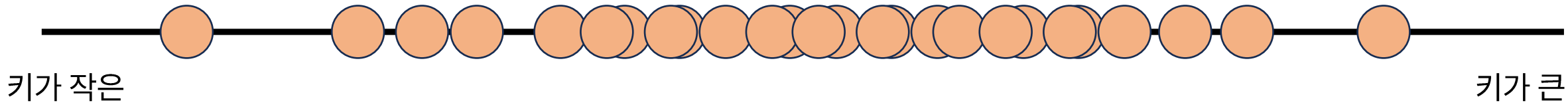
□ plotly

AI/BigData 인텐시브 과정

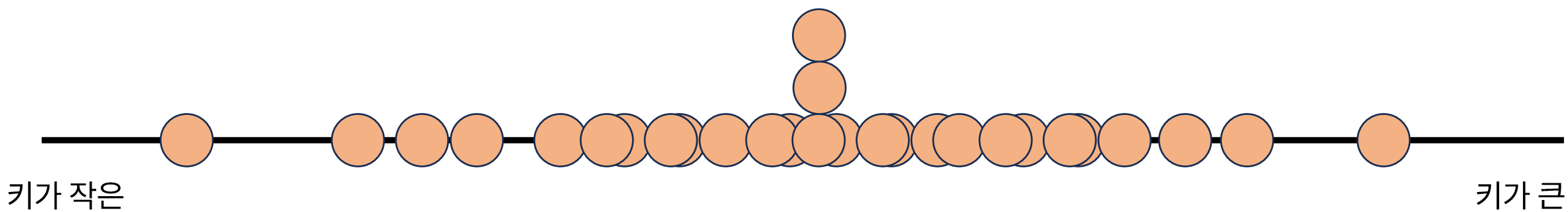
Section 1. 데이터 시각화

Section 1-1. matplotlib

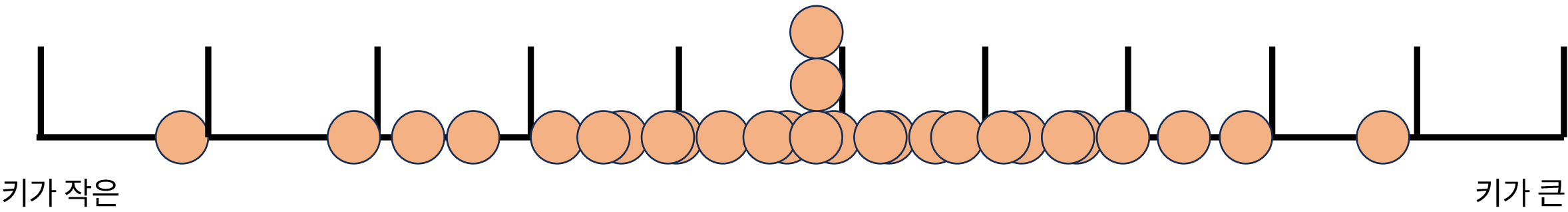
히스토그램



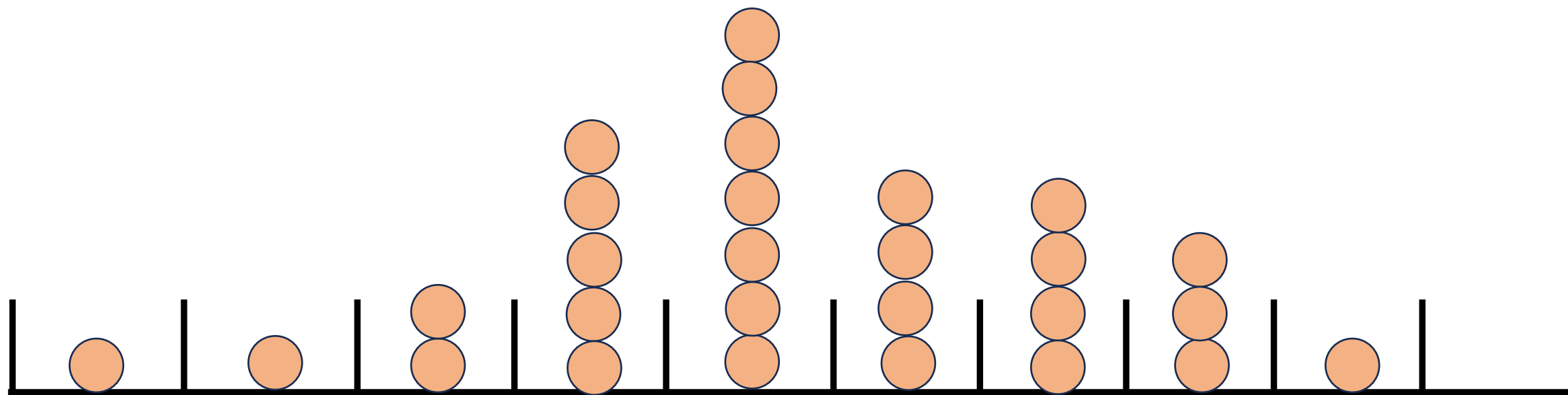
히스토그램



히스토그램



히스토그램



키가 작은

키가 큰

데이터 불러오기

```
In [1]: 1 import numpy as np
        2 import pandas as pd
```

```
In [2]: 1 df = pd.read_csv('./data/sales.csv', encoding='cp949')
```

```
In [3]: 1 df.head(5)
```

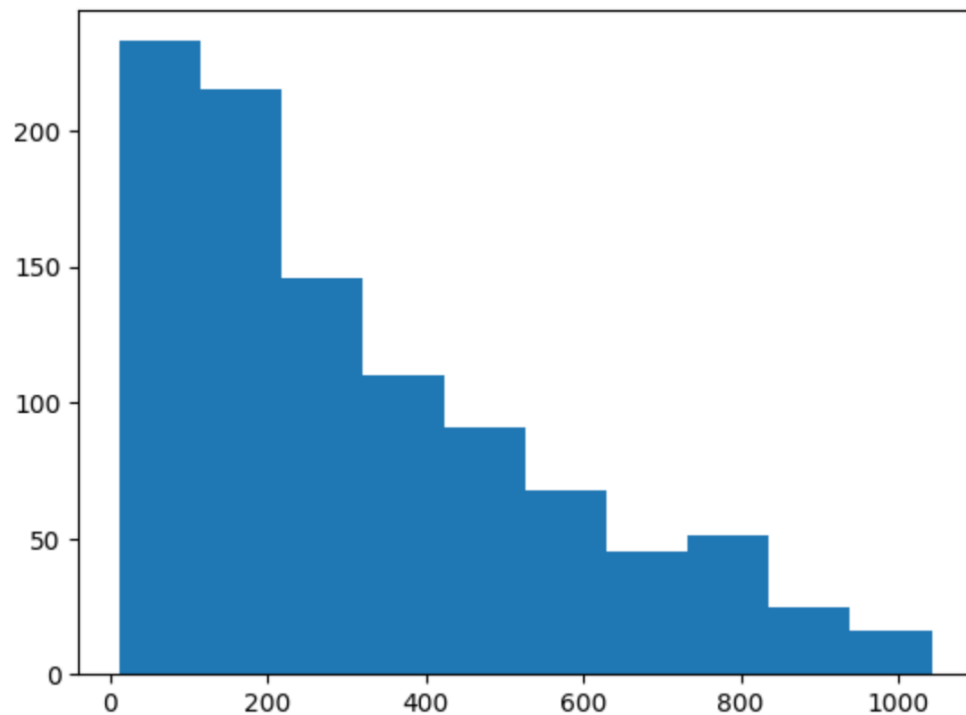
Out[3]:

	ORDERID	PURCHASE_PAGE	CITY	CUSTOMER_TYPE	GENDER	PRODUCT_TYPE	UNIT_PRICE	QUANTITY	MILEAGE	TOTAL_PRICE	DATE	PURCHASE_TIME	
0	750-67-8428	A	Seoul	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	13:08	Seoul
1	226-31-3081	C	Gyeonggi	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	10:29	
2	631-41-3108	A	Seoul	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	13:23	
3	123-19-1176	A	Seoul	Member	Male	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	20:33	Seoul
4	373-73-7910	A	Seoul	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	10:37	Seoul

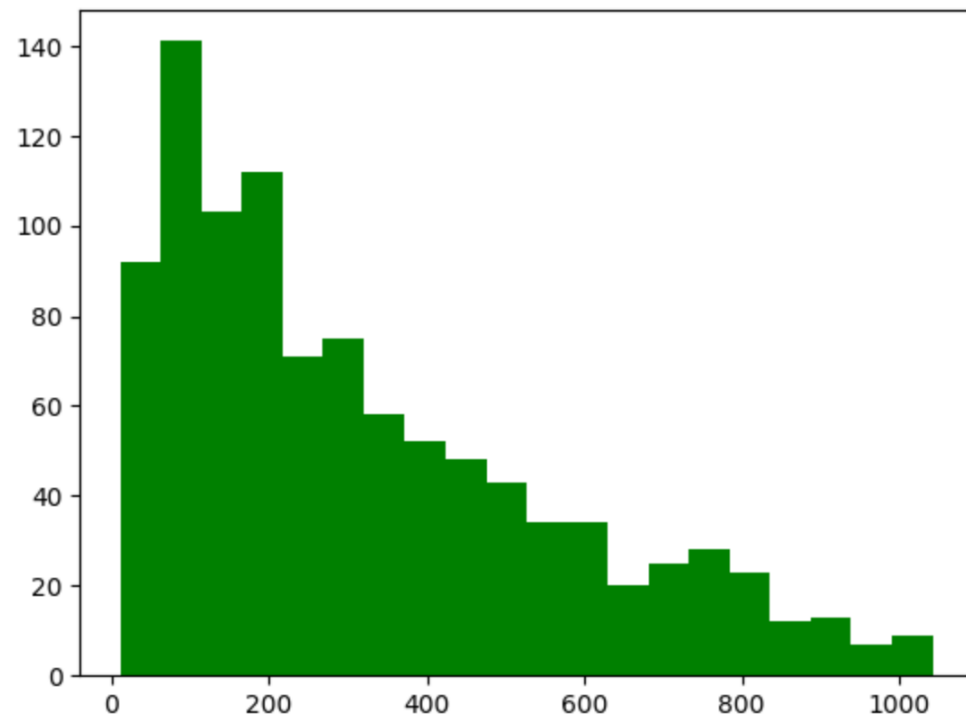
히스토그램

In [4]: ▶ 1 `import matplotlib.pyplot as plt`

In [5]: ▶ 1 `plt.hist(df['TOTAL_PRICE'])`
2 `plt.show()`



In [6]: ▶ 1 `plt.hist(df['TOTAL_PRICE'], bins=20, color='green')`
2 `plt.show()`



색상 표

https://matplotlib.org/3.1.0/gallery/color/named_colors.html

Base Colors

b	c	k
g	m	w
r	y	

Tableau Palette

tab:blue	tab:brown
tab:orange	tab:pink
tab:green	tab:gray
tab:red	tab:olive
tab:purple	tab:cyan

CSS Colors

black	bisque	forestgreen	slategrey
dimgray	darkorange	limegreen	lightsteelblue
dimgray	burlywood	darkgreen	cornflowerblue
gray	antiquewhite	green	royalblue
grey	tan	lime	ghostwhite
darkgray	navajowhite	seagreen	lavender
darkgrey	blanchedalmond	mediumseagreen	midnightblue
silver	papayawhip	springgreen	navy
lightgray	moccasin	mintcream	darkblue
lightgrey	orange	mediumspringgreen	mediumblue
gainsboro	wheat	mediumaquamarine	blue
whitesmoke	oldlace	aquamarine	slateblue
white	floralwhite	turquoise	darkslateblue
snow	darkgoldenrod	lightseagreen	mediumslateblue
rosybrown	goldenrod	mediumturquoise	mediumpurple
lightcoral	cornsilk	azure	rebeccapurple
indianred	gold	lightcyan	blueviolet
brown	lemonchiffon	paleturquoise	indigo
firebrick	khaki	darkslategray	darkorchid
maroon	palegoldenrod	darkslategrey	darkviolet
darkred	darkkhaki	teal	mediumorchid
red	ivory	darkcyan	thistle
mistyrose	beige	aqua	plum
salmon	lightyellow	cyan	violet
tomato	lightgoldenrodyellow	darkturquoise	purple
darksalmon	olive	cadetblue	darkmagenta
coral	yellow	powderblue	fuchsia
orangered	olivedrab	lightblue	magenta
lightsalmon	yellowgreen	deepskyblue	orchid
sienna	darkolivegreen	skyblue	mediumvioletred
seashell	greenyellow	lightskyblue	deeppink
chocolate	chartreuse	steelblue	hotpink
saddlebrown	lawngreen	aliceblue	lavenderblush
sandybrown	honeydew	dodgerblue	palevioletred
peachpuff	darkseagreen	lightslategray	crimson
peru	palegreen	lightslategrey	pink
linen	lightgreen	slategray	lightpink

Section

matplotlib

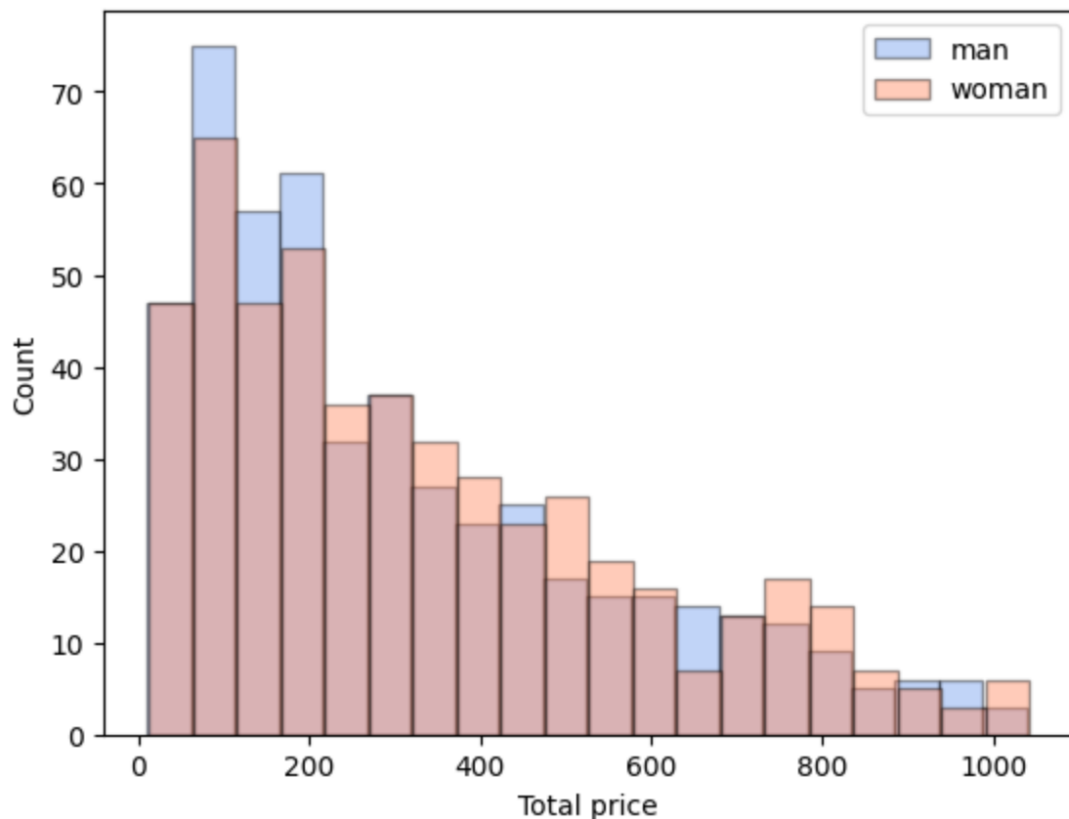
```
In [7]: 1 man_price = df[df['GENDER']=='Male']['TOTAL_PRICE']  
2 woman_price = df[df['GENDER']=='Female']['TOTAL_PRICE']
```

```
In [8]: 1 plt.hist(man_price, bins=20, color='cornflowerblue', alpha=0.4, edgecolor='black', linewidth=1, label='man')  
2 plt.hist(woman_price, bins=20, color='coral', alpha=0.4, edgecolor='black', linewidth=1, label='woman')  
3 plt.xlabel('Total price')  
4 plt.ylabel('Count')  
5 plt.legend()  
6 plt.show()
```

투명도

테두리 색깔

테두리 두께



막대 그래프(barplot)

```
In [44]: ▶ 1 category_counts = df['CITY'].value_counts()  
          2 print(category_counts)
```

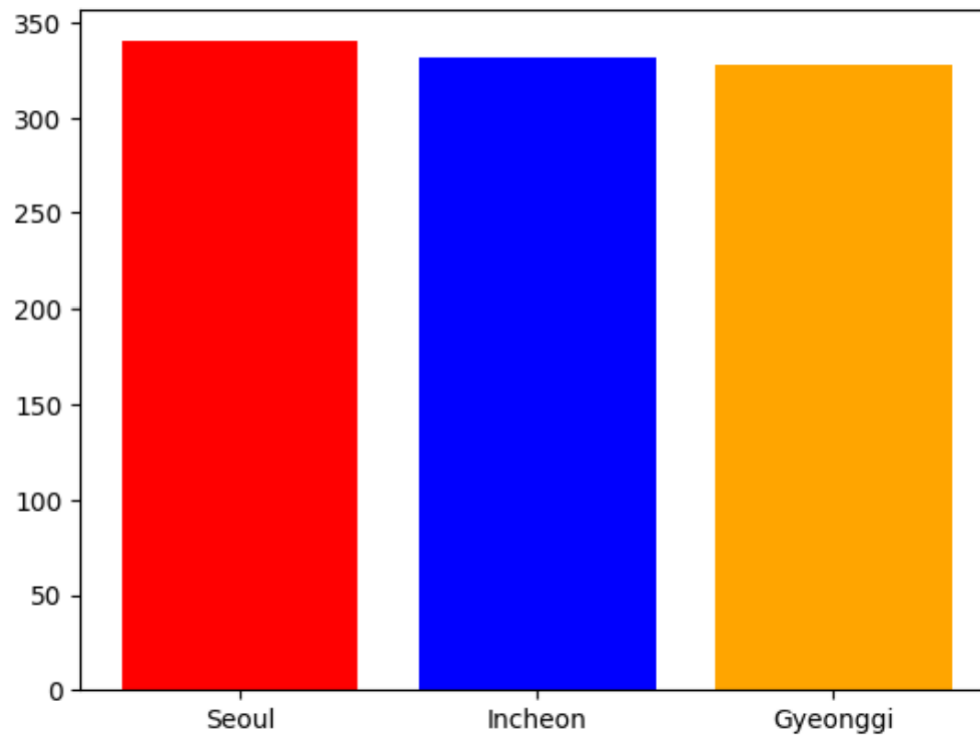
```
Seoul      340  
Incheon    332  
Gyeonggi   328  
Name: CITY, dtype: int64
```

```
In [45]: ▶ 1 city_idx = category_counts.index  
          2 print(city_idx)
```

```
Index(['Seoul', 'Incheon', 'Gyeonggi'], dtype='object')
```

```
In [49]: ▶ 1 colors = ['red', 'blue', 'orange']
```

```
In [50]: ▶ 1 plt.bar(x=city_idx, height=category_counts, color=colors)  
          2 plt.show()
```



막대 그래프(barplot)

```
In [53]: 1 city_gender = pd.crosstab(df['CITY'], df['GENDER'])  
2 city_gender
```

Out[53]:

CITY	GENDER	
	Female	Male
Gyeonggi	178	150
Incheon	162	170
Seoul	161	179

```
In [57]: 1 city_gender.index
```

Out[57]: Index(['Gyeonggi', 'Incheon', 'Seoul'], dtype='object', name='CITY')

```
In [58]: 1 city_gender.columns
```

Out[58]: Index(['Female', 'Male'], dtype='object', name='GENDER')

```
In [61]: 1 city_gender['Male']
```

Out[61]: CITY
Gyeonggi 150
Incheon 170
Seoul 179
Name: Male, dtype: int64

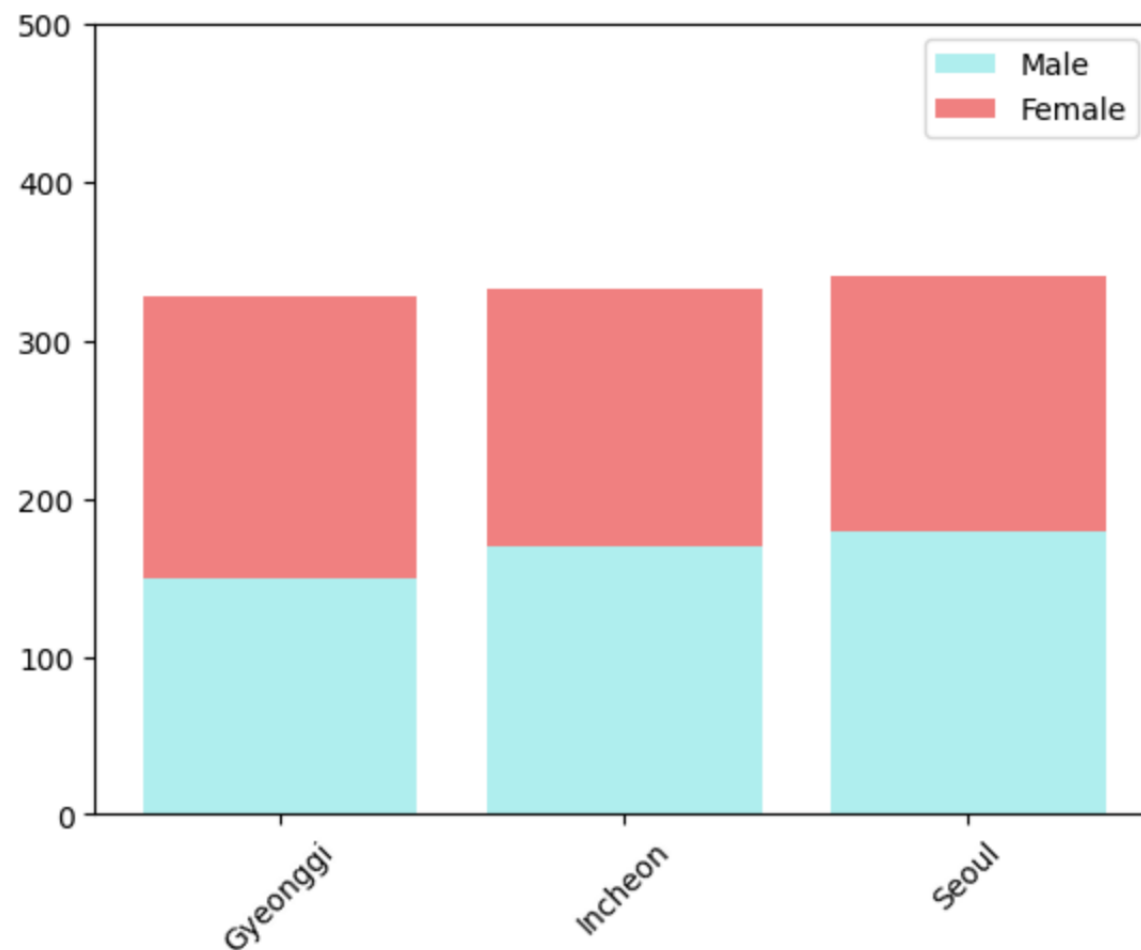
```
In [59]: 1 city_gender['Female']
```

Out[59]: CITY
Gyeonggi 178
Incheon 162
Seoul 161
Name: Female, dtype: int64

Section matplotlib

In [80]: ▶

```
1 plt.bar(x=city_gender.index, height=city_gender['Male'], color='paleturquoise', label='Male')
2 plt.bar(x=city_gender.index, height=city_gender['Female'], bottom=city_gender['Male'],
3         color='lightcoral', label='Female')
4 plt.ylim(0, 500)
5 plt.xticks(rotation=45)
6 plt.legend()
7 plt.show()
```



In [19]: ▶

```
1 city_price = df.groupby('CITY')['TOTAL_PRICE'].mean()  
2 city_price
```

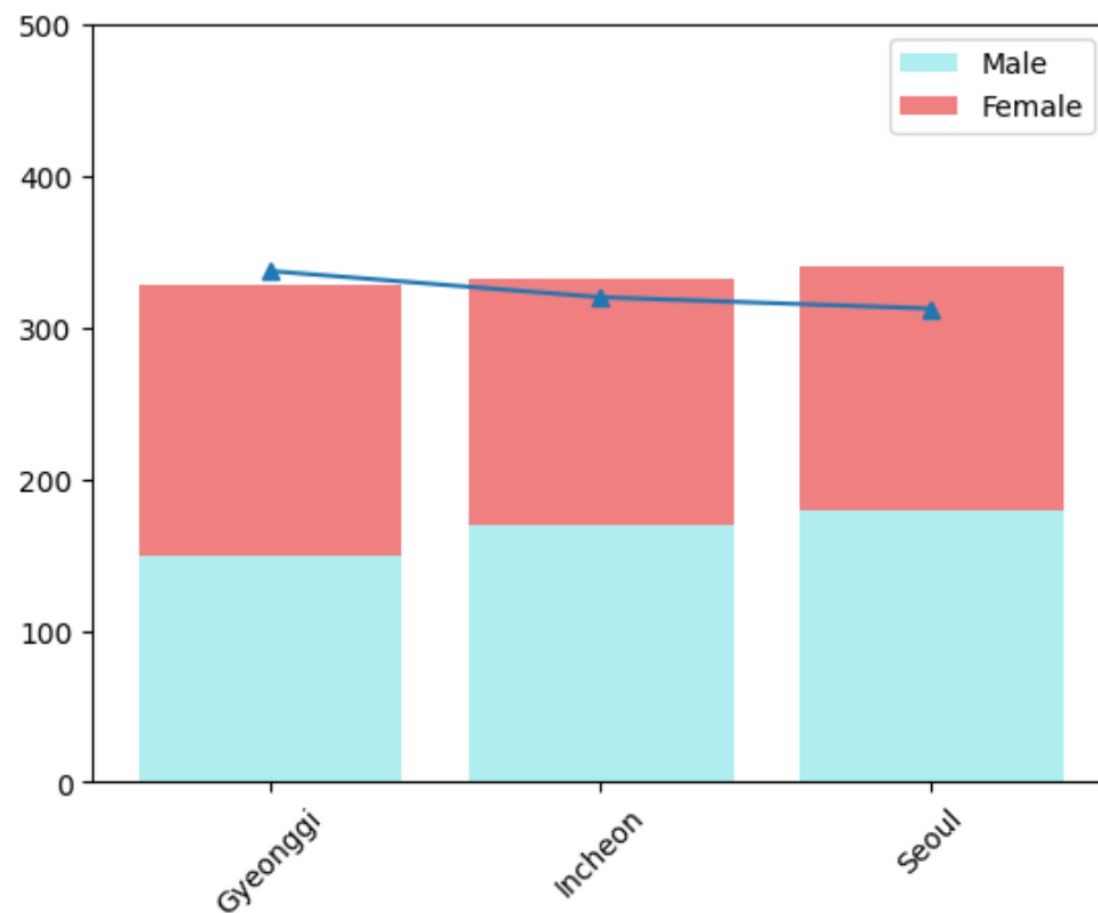
Out[19]:

CITY	
Gyeonggi	337.099715
Incheon	319.872506
Seoul	312.354031

Name: TOTAL_PRICE, dtype: float64

Section matplotlib

```
In [68]: ▶ 1 plt.bar(x=city_gender.index, height=city_gender['Male'], color='paleturquoise', label='Male')
2 plt.bar(x=city_gender.index, height=city_gender['Female'], bottom=city_gender['Male'],
3         color='lightcoral', label='Female')
4 plt.plot(city_gender.index, city_price, '^-.')
5 plt.ylim(0, 500)
6 plt.xticks(rotation=45)
7 plt.legend()
8 plt.show()
```

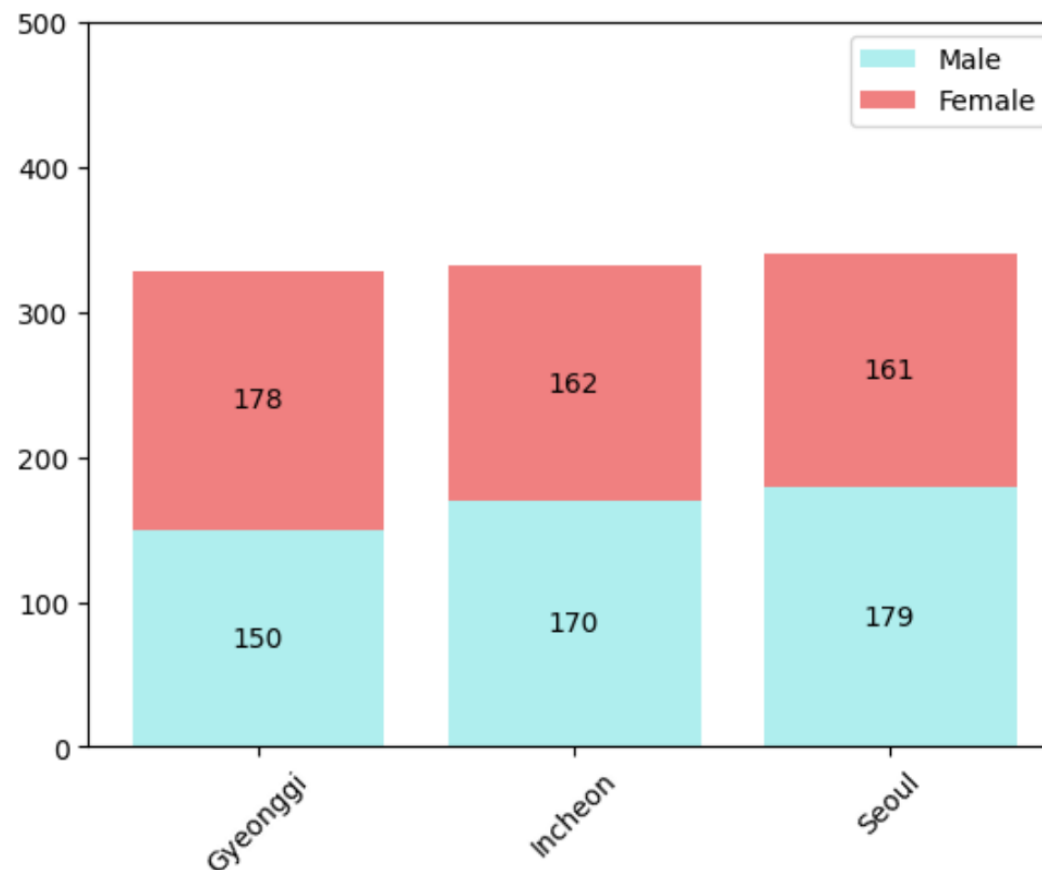


Section

matplotlib

In [85]:

```
1 fig, ax = plt.subplots()
2 bar1 = ax.bar(x=city_gender.index, height=city_gender['Male'], color='paleturquoise', label='Male')
3 bar2 = ax.bar(x=city_gender.index, height=city_gender['Female'], bottom=city_gender['Male'],
4               color='lightcoral', label='Female')
5 ax.bar_label(bar1, label_type='center')
6 ax.bar_label(bar2, label_type='center')
7 plt.ylim(0, 500)
8 plt.legend()
9 plt.xticks(rotation=45)
10 plt.show()
```



비율 막대 그래프(barplot)

```
In [105]: 1 city_gender2 = pd.crosstab(df['CITY'], df['GENDER'], normalize='index')
          2 city_gender2 = city_gender2.round(decimals=3)*100
          3 city_gender2
```

Out[105]:

CITY	GENDER	
	Female	Male
Gyeonggi	54.3	45.7
Incheon	48.8	51.2
Seoul	47.4	52.6

```
In [106]: 1 city_gender2.index
```

Out[106]: Index(['Gyeonggi', 'Incheon', 'Seoul'], dtype='object', name='CITY')

```
In [107]: 1 city_gender2.columns
```

Out[107]: Index(['Female', 'Male'], dtype='object', name='GENDER')

```
In [108]: 1 city_gender2['Male']
```

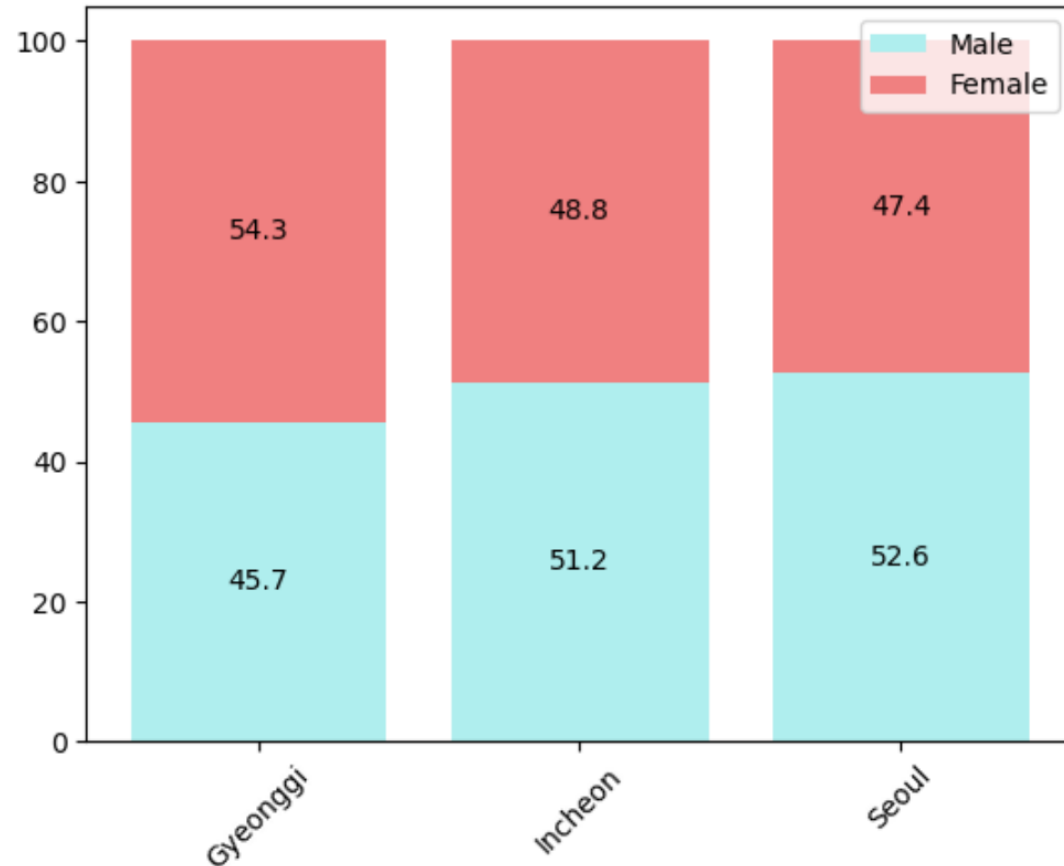
Out[108]: CITY
Gyeonggi 45.7
Incheon 51.2
Seoul 52.6
Name: Male, dtype: float64

```
In [109]: 1 city_gender2['Female']
```

Out[109]: CITY
Gyeonggi 54.3
Incheon 48.8
Seoul 47.4
Name: Female, dtype: float64

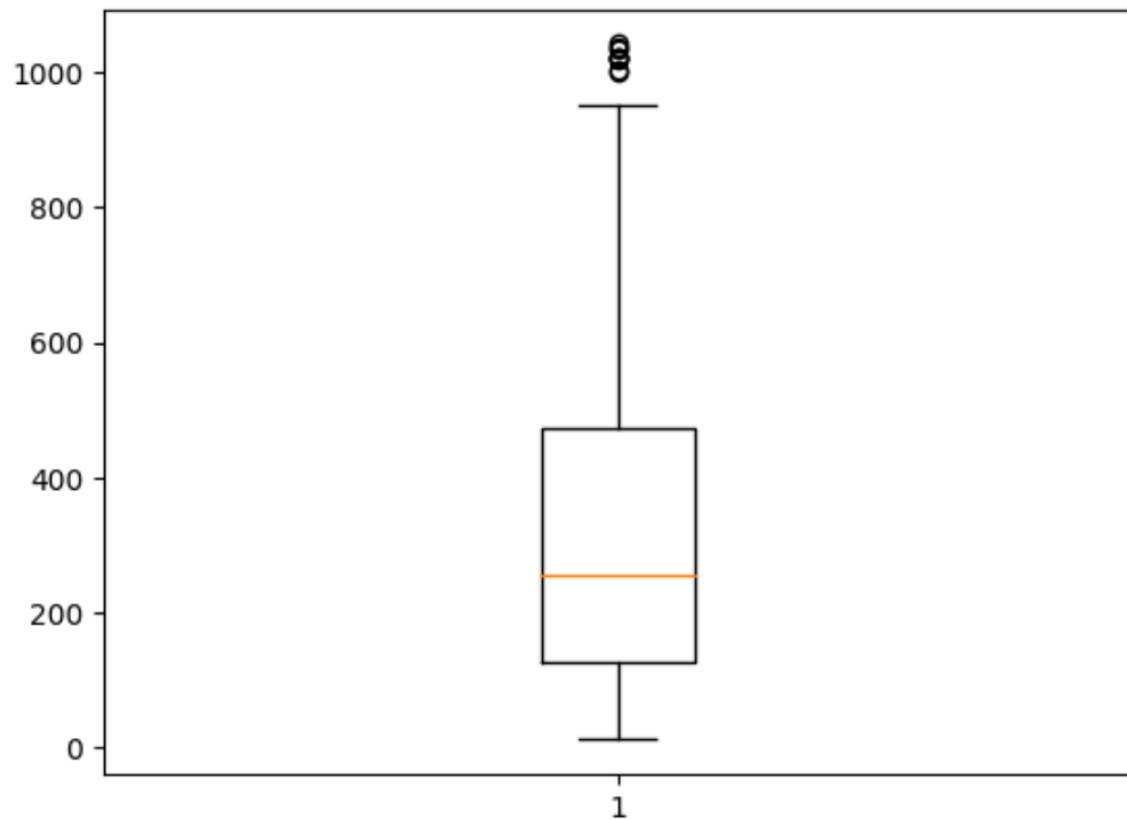
Section matplotlib

```
In [110]: ▶ 1 fig, ax = plt.subplots()
2 bar3 = ax.bar(x=city_gender2.index, height=city_gender2['Male'], color='paleturquoise', label='Male')
3 bar4 = ax.bar(x=city_gender2.index, height=city_gender2['Female'], bottom=city_gender2['Male'],
4               color='lightcoral', label='Female')
5 ax.bar_label(bar3, label_type='center')
6 ax.bar_label(bar4, label_type='center')
7 plt.legend()
8 plt.xticks(rotation=45)
9 plt.show()
```

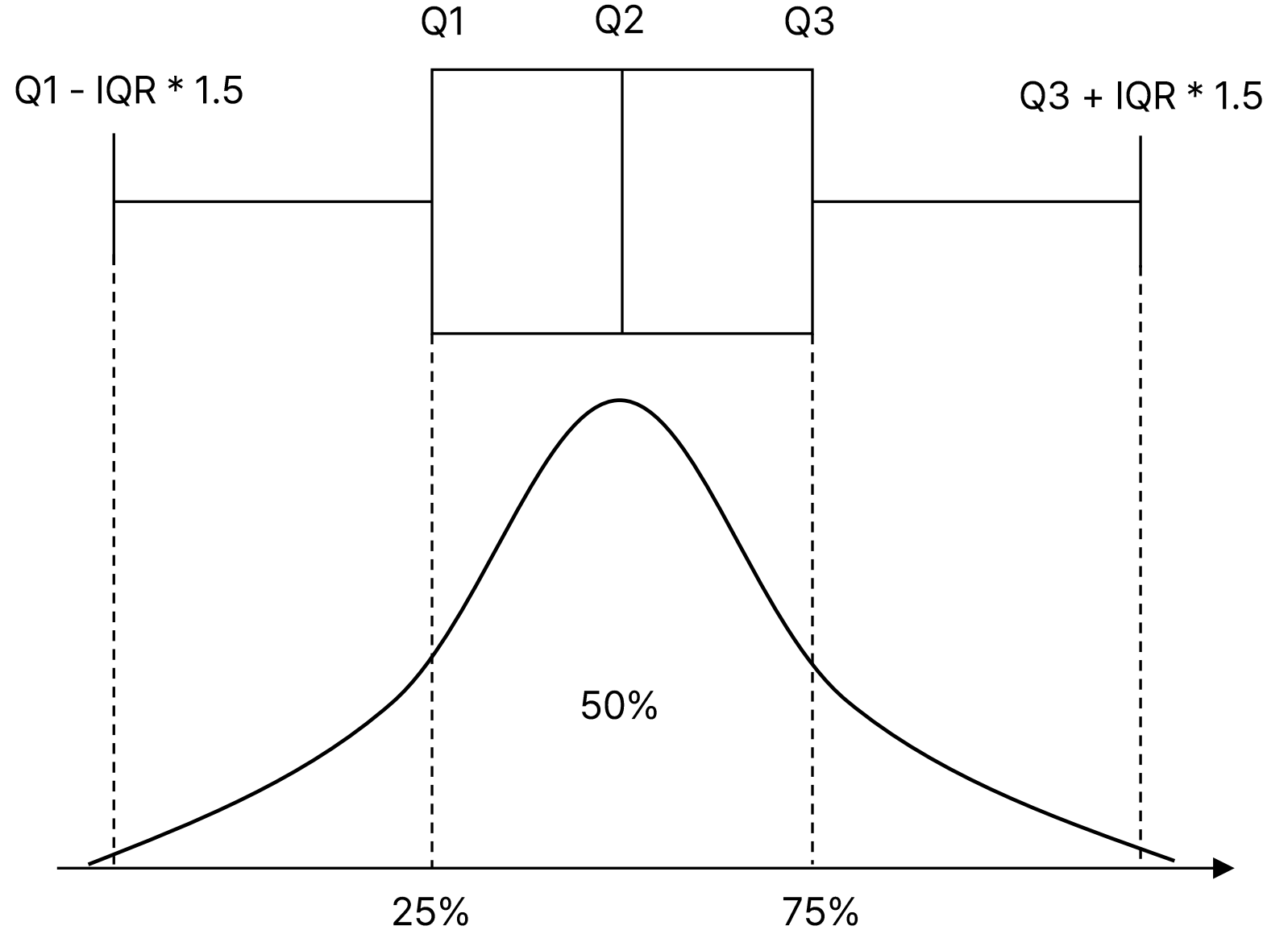


상자 그림(boxplot)

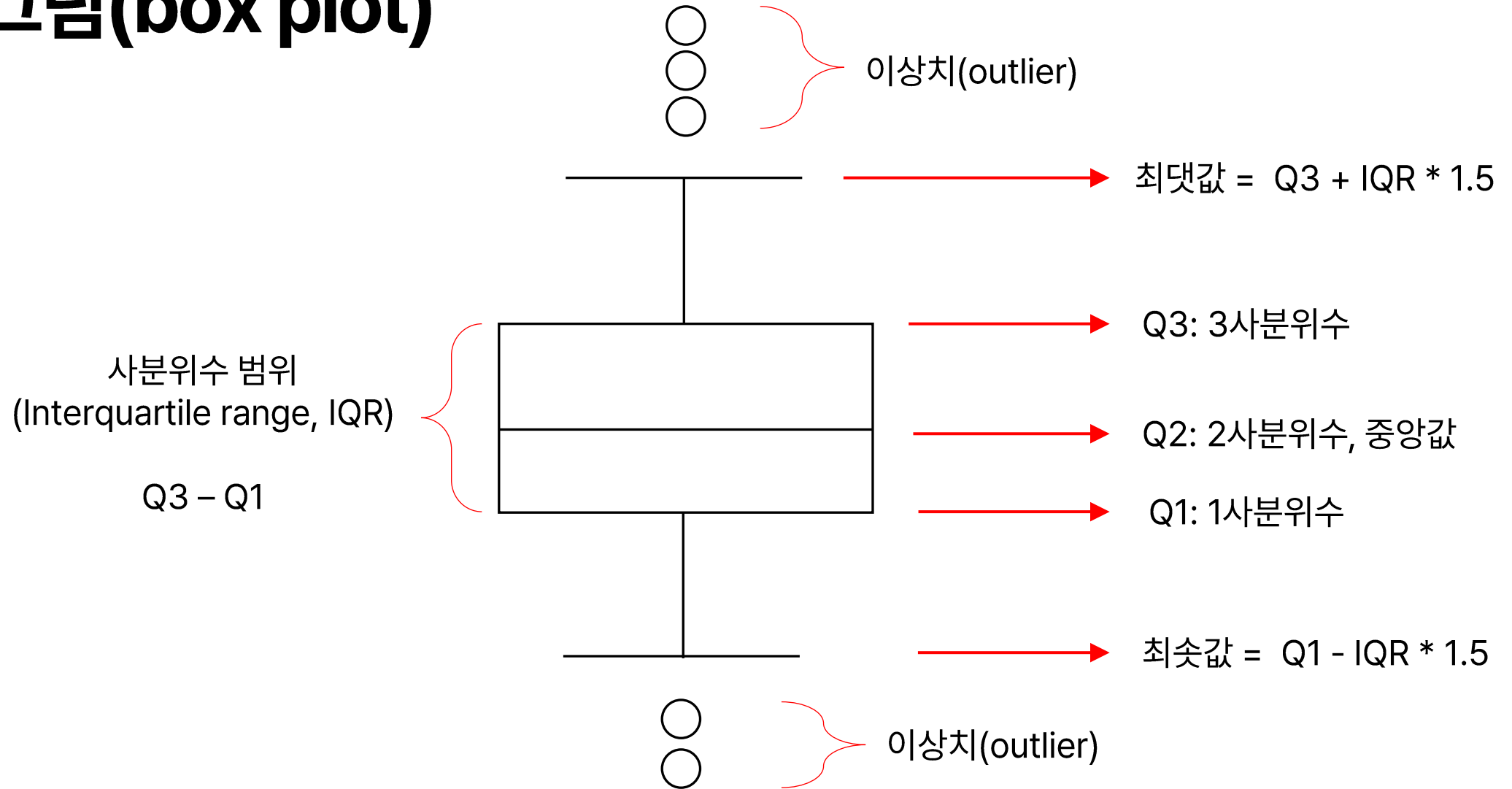
```
In [9]: ▶ 1 plt.boxplot(df['TOTAL_PRICE'])  
        2 plt.show()
```



상자 그림(boxplot)

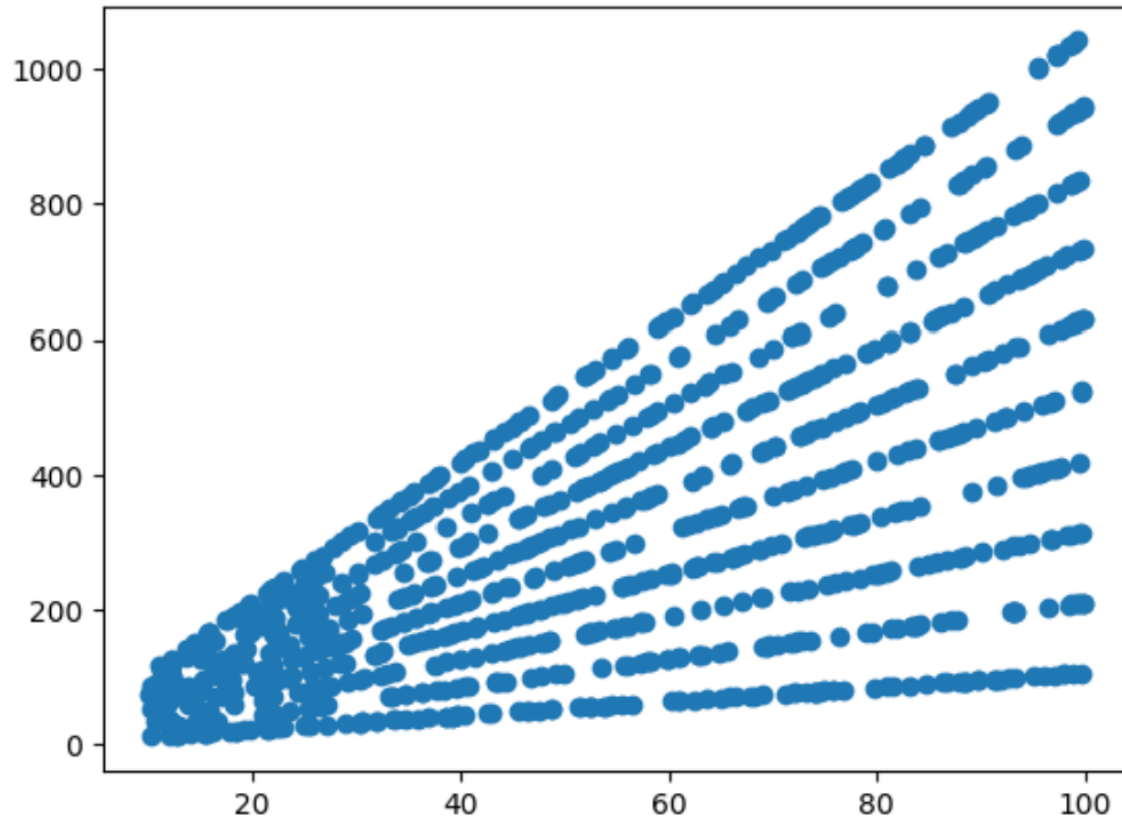


상자 그림(box plot)



산점도(scatter plot)

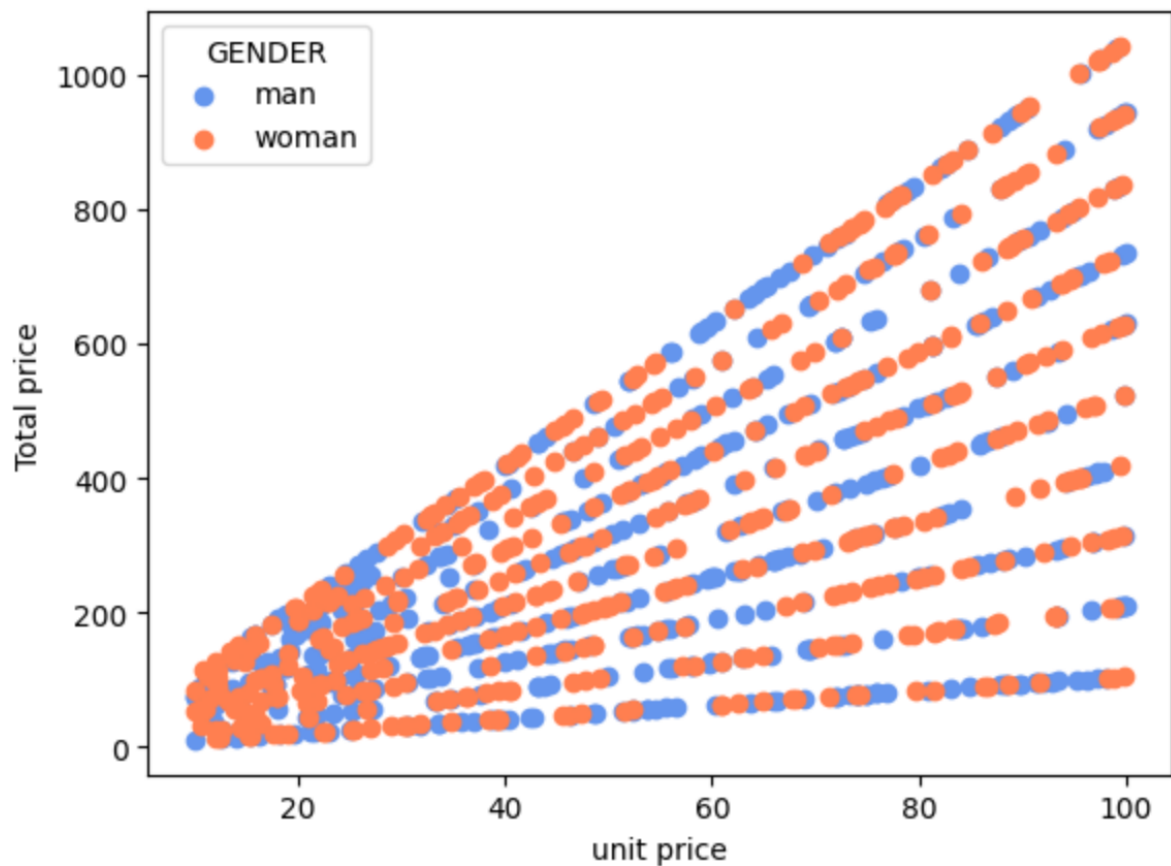
```
In [10]: ▶ 1 plt.scatter(df['UNIT_PRICE'], df['TOTAL_PRICE'])  
2 plt.show()
```



Section matplotlib

```
In [11]: ▶ 1 man = df[df['GENDER']=='Male']  
          2 woman = df[df['GENDER']=='Female']
```

```
In [12]: ▶ 1 plt.scatter(man['UNIT_PRICE'], man['TOTAL_PRICE'], color='cornflowerblue', label='man')  
          2 plt.scatter(woman['UNIT_PRICE'], woman['TOTAL_PRICE'], color='coral', label='woman')  
          3 plt.legend(title='GENDER')  
          4 plt.xlabel('unit price')  
          5 plt.ylabel('Total price')  
          6 plt.show()
```



파이 차트(pie chart)

In [13]: ▶

```
1 category_counts = df['CITY'].value_counts()  
2 print(category_counts)
```

```
Seoul      340  
Incheon    332  
Gyeonggi   328  
Name: CITY, dtype: int64
```

도시별 빈도수 확인

In [14]: ▶

```
1 city_idx = category_counts.index  
2 print(city_idx)
```

```
Index(['Seoul', 'Incheon', 'Gyeonggi'], dtype='object')
```

도시명 확인

In [15]: ▶

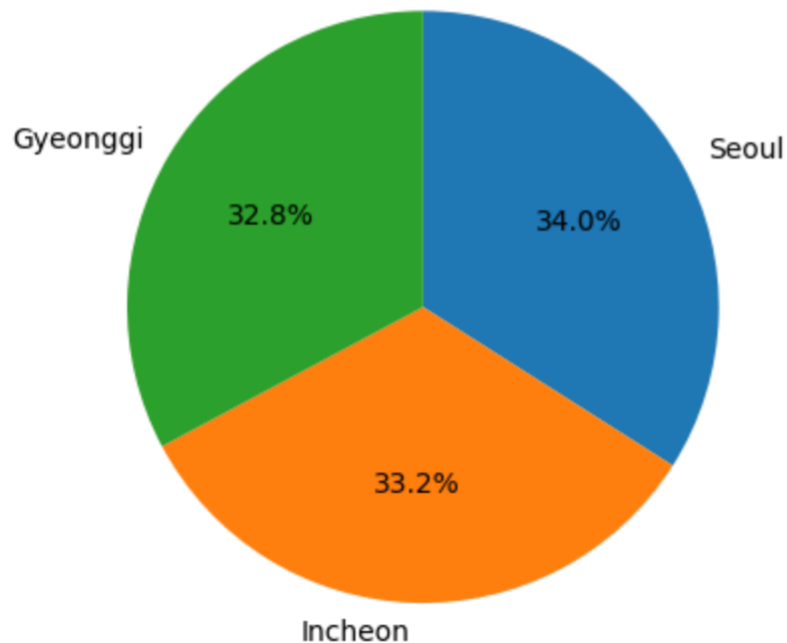
```
1 plt.pie(category_counts, labels=city_idx, autopct='%.1f%%', startangle=90, counterclock=False)  
2 plt.show()
```

부채꼴 내 숫자표기

.1f -> 소수점 첫째 자리까지 표기
.2f -> 소수점 둘째 자리까지 표기

부채꼴 시작 각도

반시계방향 여부
False => 시계방향
True => 반시계방향



파이 차트(pie chart)

In [16]: ▶

```
1 city_list = list(category_counts.index)
2 expl = []
3 for city in city_list:
4     if city=='Seoul':
5         expl.append(0.1)
6     else:
7         expl.append(0)
```

→ 부채꼴과 중심까지의 거리

→ 'Seoul' 부채꼴과 중심과의 거리 0.1

→ 'Seoul' 이 아닌 나머지 부채꼴은 중심과의 거리 0
즉, Seoul이 아닌 나머지 부채꼴은 중심과 붙어있음

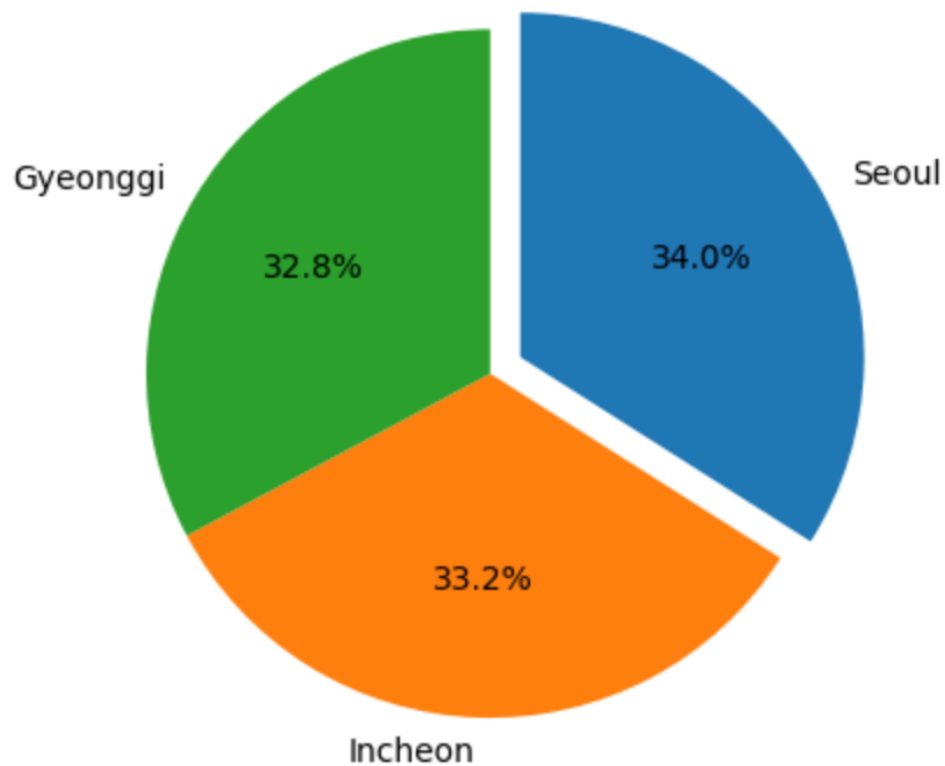
In [17]: ▶

```
1 expl
```

Out[17]: [0.1, 0, 0]

파이 차트(pie chart)

```
In [18]: ▶ 1 plt.pie(category_counts, labels=city_idx, autopct='%.1f%%', startangle=90, counterclock=False, explode=explode)
          2 plt.show()
```

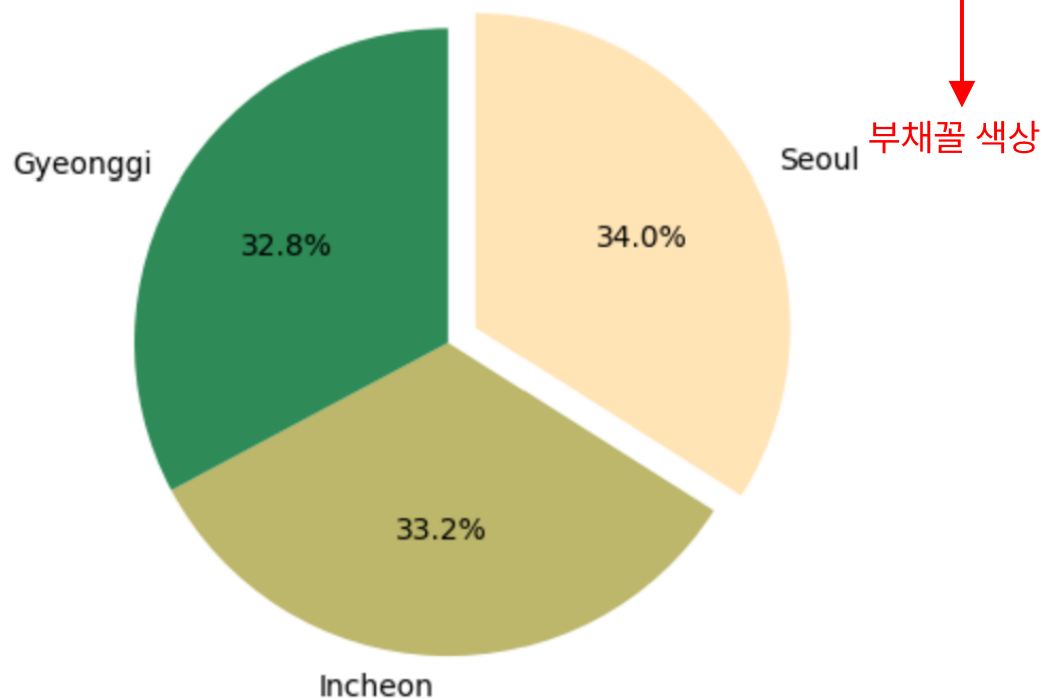


부채꼴과 중심까지의 거리

파이 차트(pie chart)

```
In [19]: ▶ 1 pie_color = ['moccasin', 'darkkhaki', 'seagreen']  
                서울      인천      경기
```

```
In [20]: ▶ 1 plt.pie(category_counts, labels=city_idx, autopct='%0.1f%%', startangle=90,  
2               counterclock=False, explode=explode, colors=pie_color)  
3 plt.show()
```



AI/BigData 인텐시브 과정

Section 1. 데이터 시각화

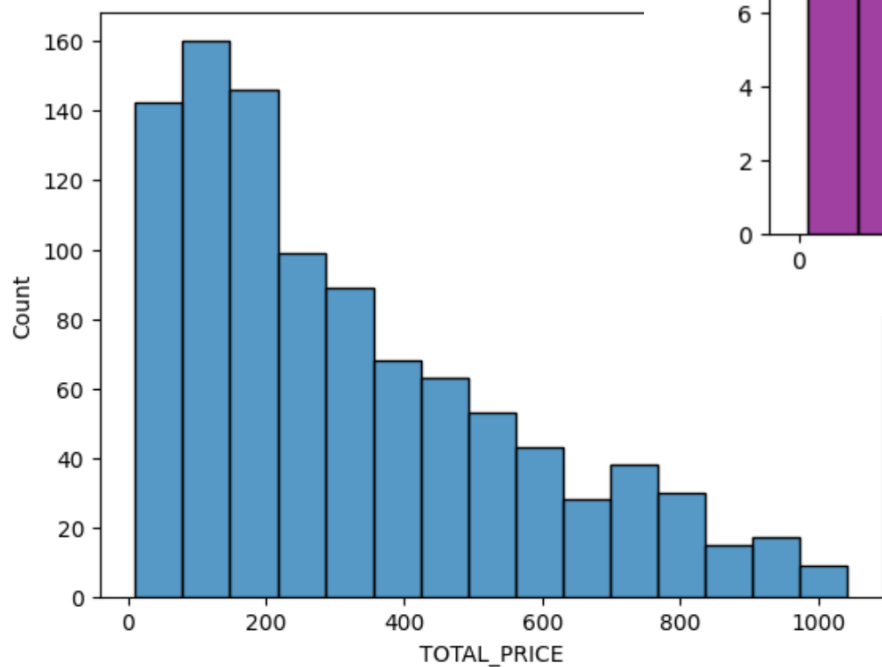
Section 1-2. seaborn

히스토그램

```
1 import seaborn as sns
```

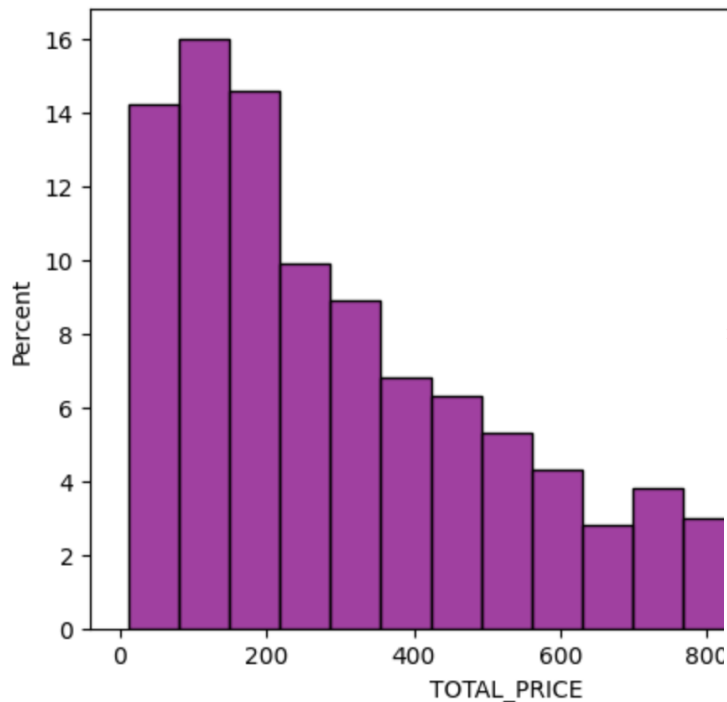
```
1 sns.histplot(data=df, x='TOTAL_PRICE')
```

<Axes: xlabel='TOTAL_PRICE', ylabel='Count'>



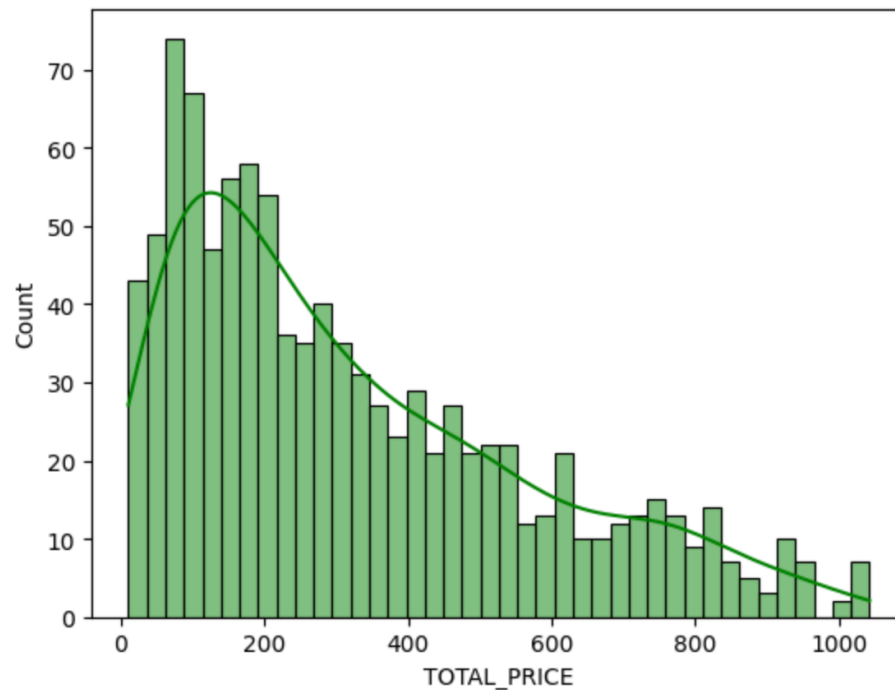
```
1 sns.histplot(data=df, x='TOTAL_PRICE', stat='percent', color='purple')
```

<Axes: xlabel='TOTAL_PRICE', ylabel='Percent'>



```
1 sns.histplot(data=df, x='TOTAL_PRICE', color='green', bins=40, kde=True)
```

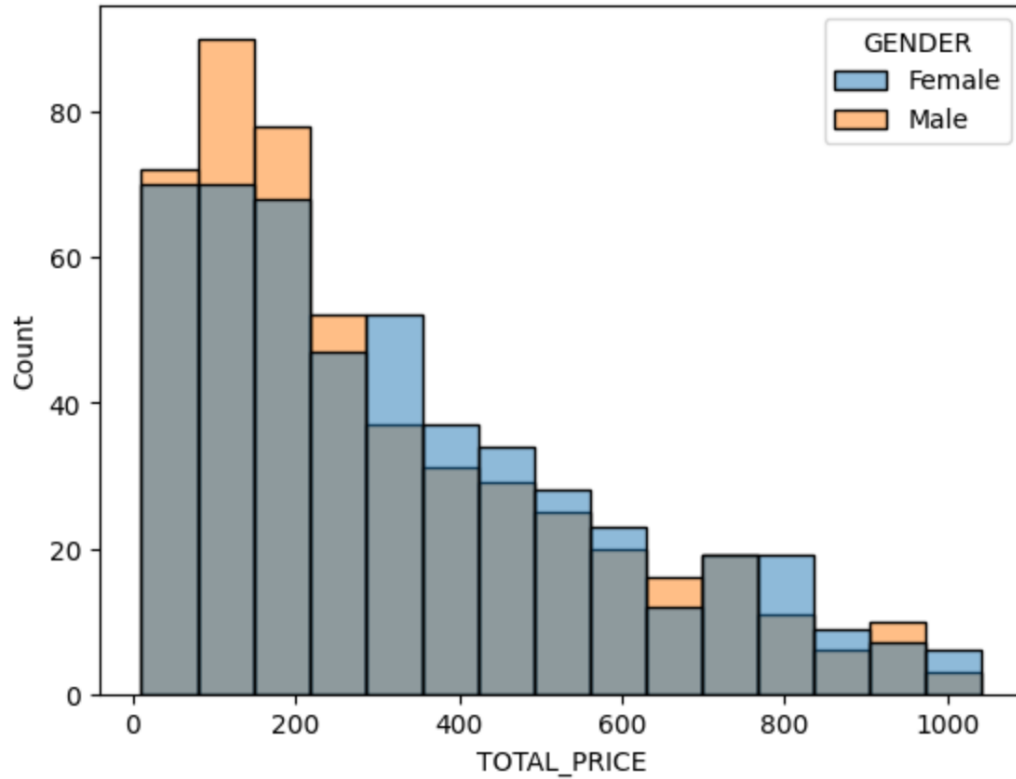
<Axes: xlabel='TOTAL_PRICE', ylabel='Count'>



히스토그램

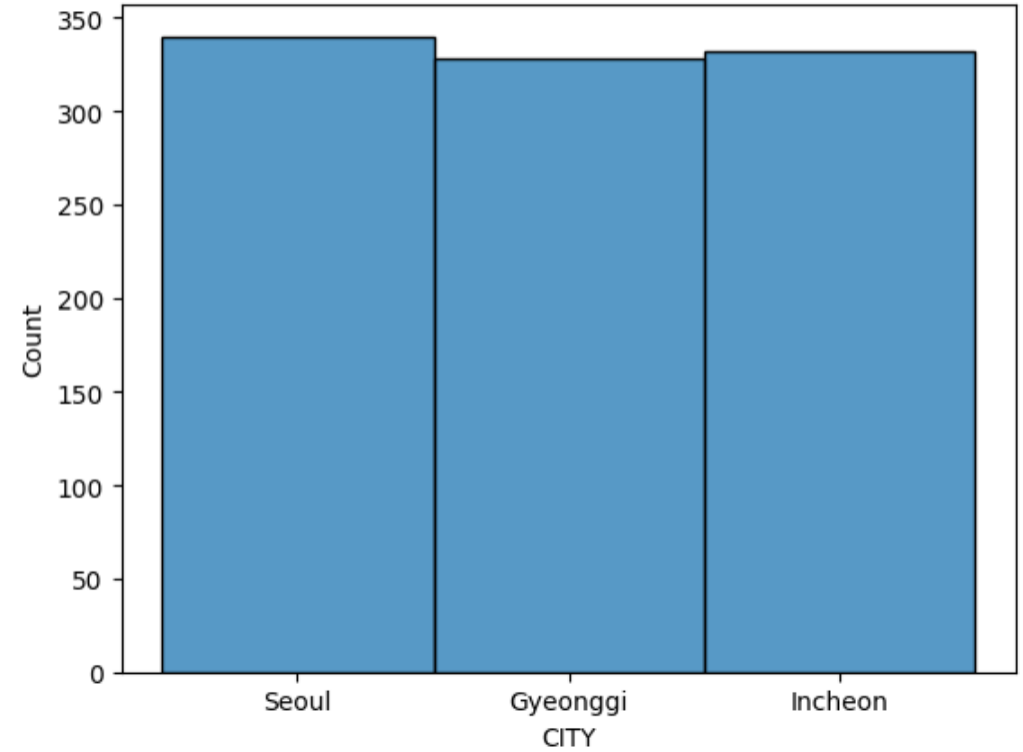
```
In [25]: 1 sns.histplot(data=df, x='TOTAL_PRICE', hue='GENDER')
```

```
Out[25]: <Axes: xlabel='TOTAL_PRICE', ylabel='Count'>
```



```
In [74]: 1 sns.histplot(data=df, x='CITY')
```

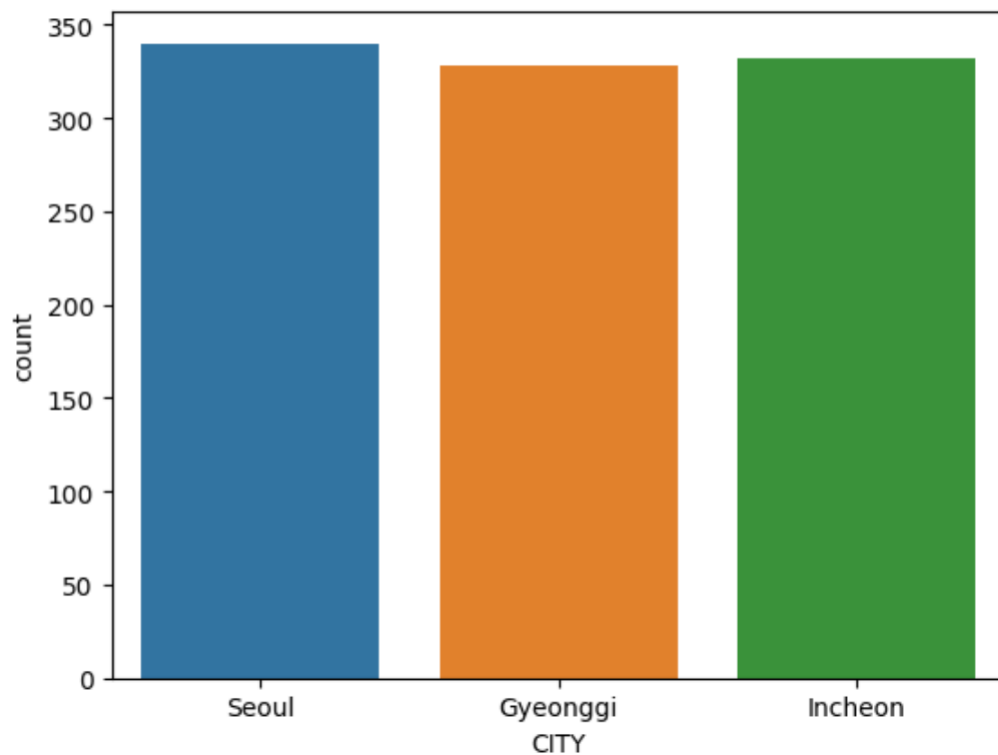
```
Out[74]: <Axes: xlabel='CITY', ylabel='Count'>
```



countplot

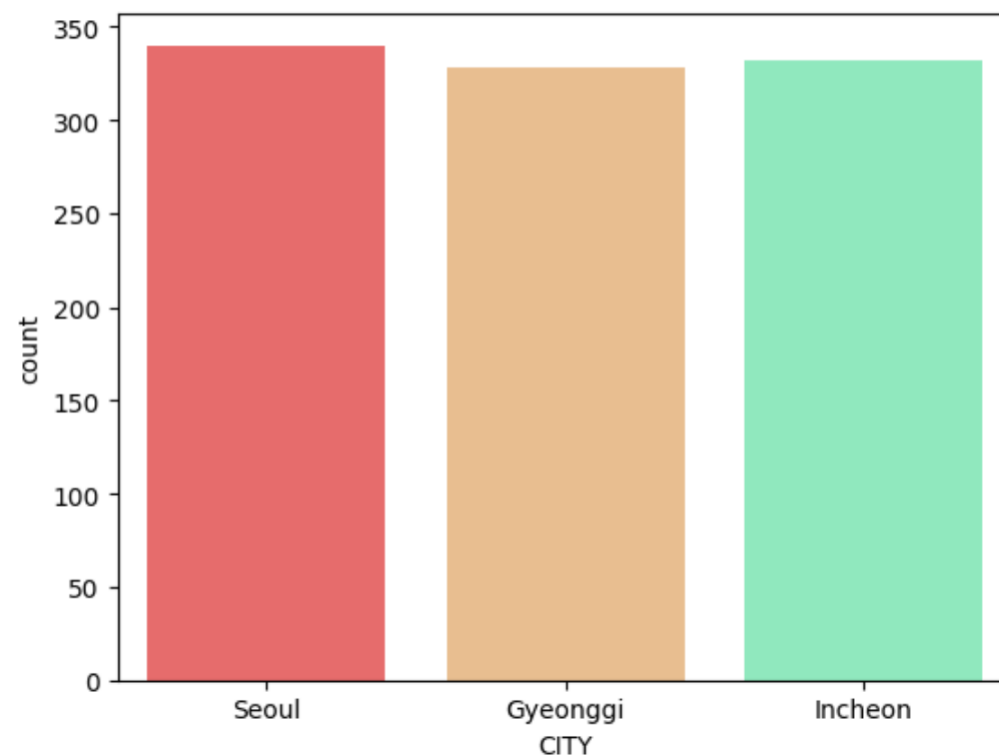
```
In [63]: 1 sns.countplot(data=df, x="CITY")
```

```
Out[63]: <Axes: xlabel='CITY', ylabel='count'>
```



```
In [71]: 1 colors = ["#FA5858", "#F7BE81", "#81F7BE"]  
2 palette1 = sns.color_palette(colors)  
3 sns.countplot(data=df, x="CITY", palette=palette1)
```

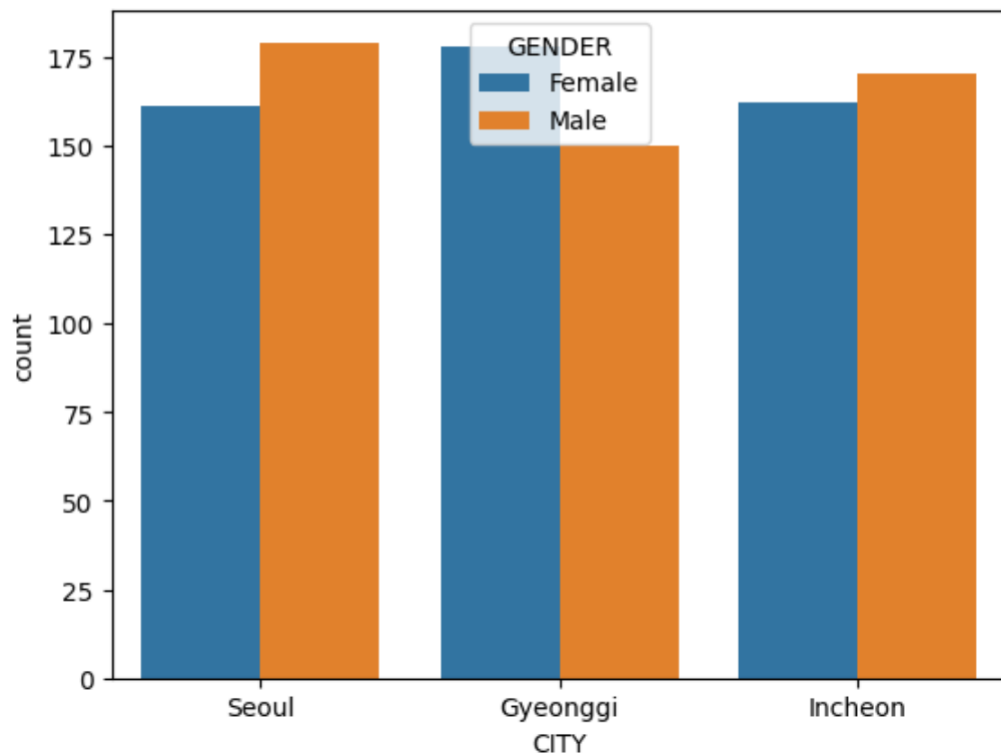
```
Out[71]: <Axes: xlabel='CITY', ylabel='count'>
```



countplot

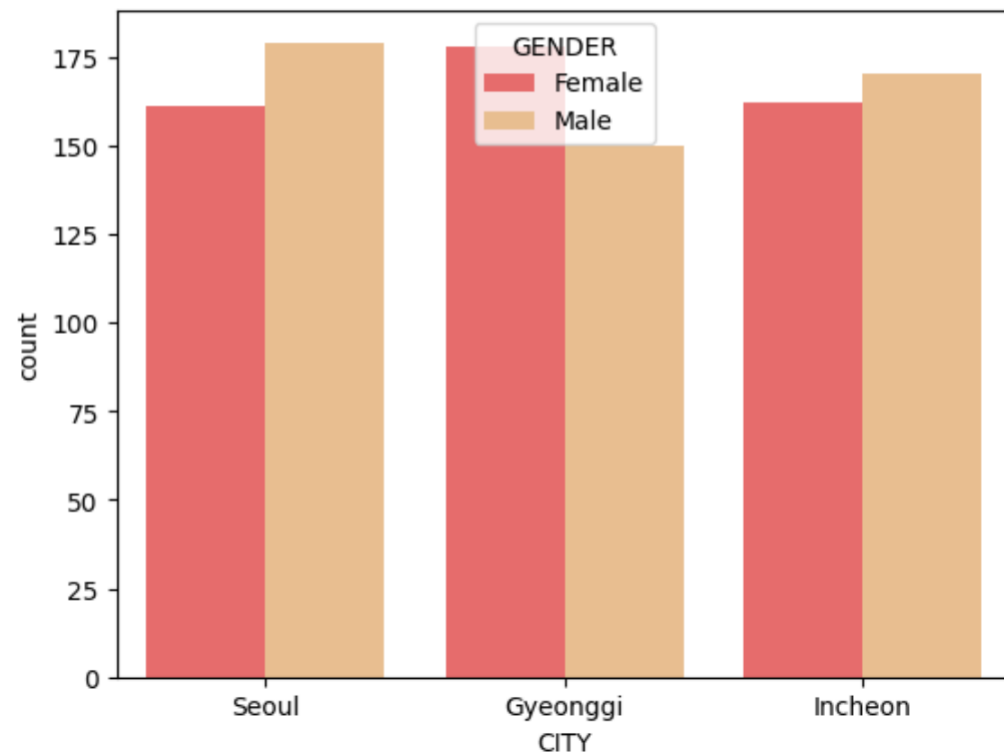
```
In [79]: 1 sns.countplot(data=df, x="CITY", hue='GENDER')
```

Out[79]: <Axes: xlabel='CITY', ylabel='count'>



```
In [77]: 1 colors = ["#FA5858", "#F7BE81", "#81F7BE"]  
2 palette1 = sns.color_palette(colors)  
3 sns.countplot(data=df, x="CITY", hue='GENDER', palette=palette1)
```

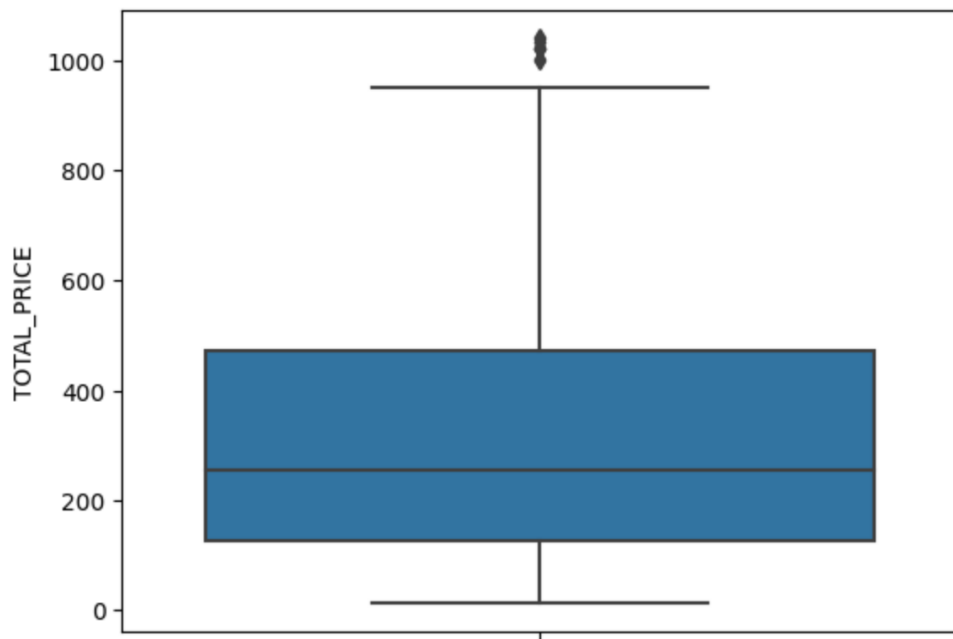
Out[77]: <Axes: xlabel='CITY', ylabel='count'>



상자 그림(boxplot)

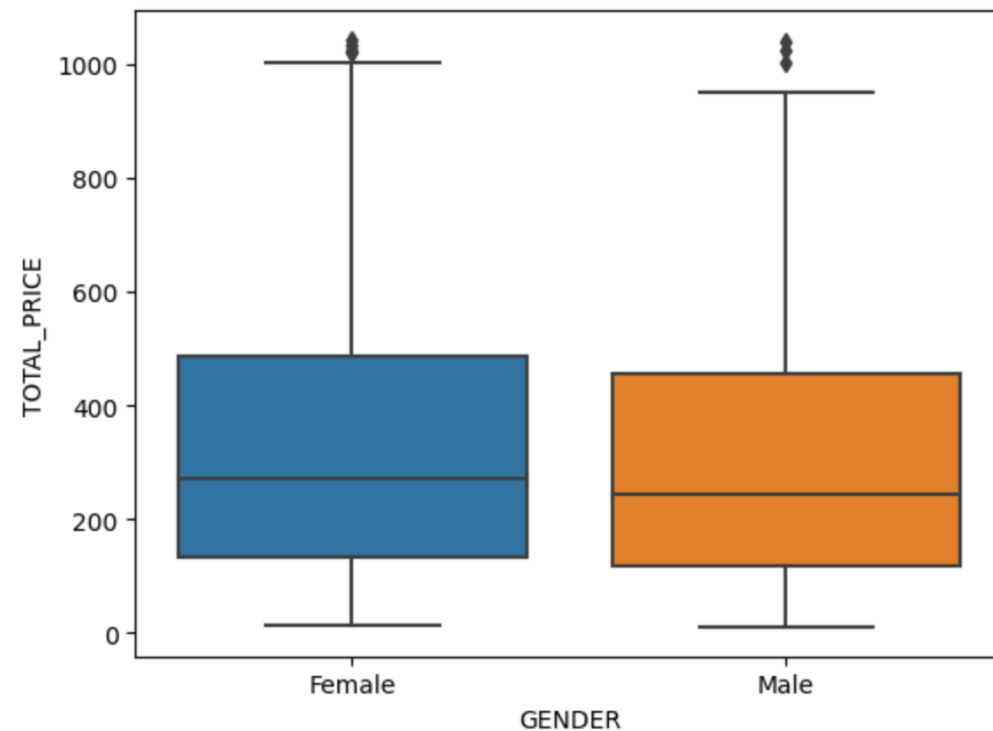
```
In [26]: 1 sns.boxplot(data=df, y='TOTAL_PRICE')
```

```
Out[26]: <Axes: ylabel='TOTAL_PRICE'>
```



```
In [27]: 1 sns.boxplot(data=df, x='GENDER', y='TOTAL_PRICE')
```

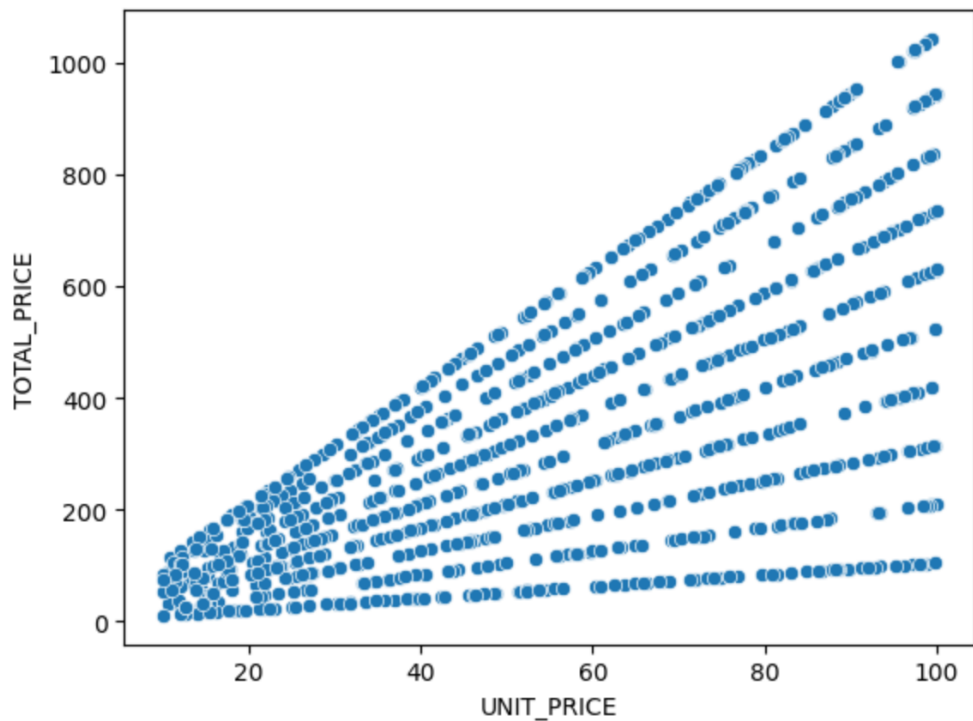
```
Out[27]: <Axes: xlabel='GENDER', ylabel='TOTAL_PRICE'>
```



산점도

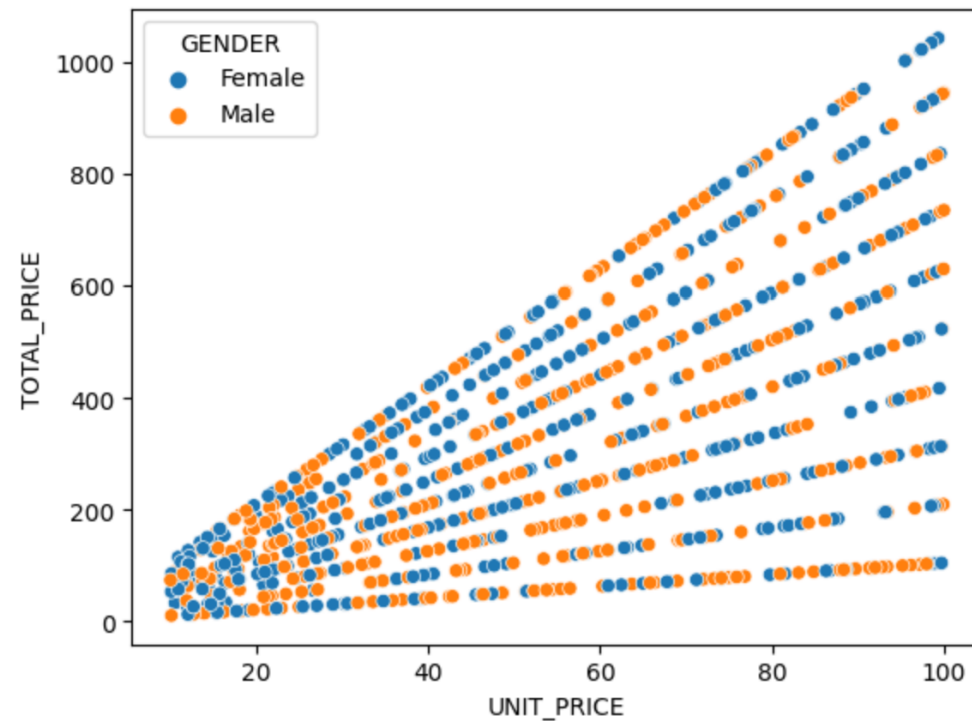
```
In [28]: 1 sns.scatterplot(data=df, x='UNIT_PRICE', y='TOTAL_PRICE')
```

```
Out[28]: <Axes: xlabel='UNIT_PRICE', ylabel='TOTAL_PRICE'>
```



```
In [29]: 1 sns.scatterplot(data=df, x='UNIT_PRICE', y='TOTAL_PRICE', hue='GENDER')
```

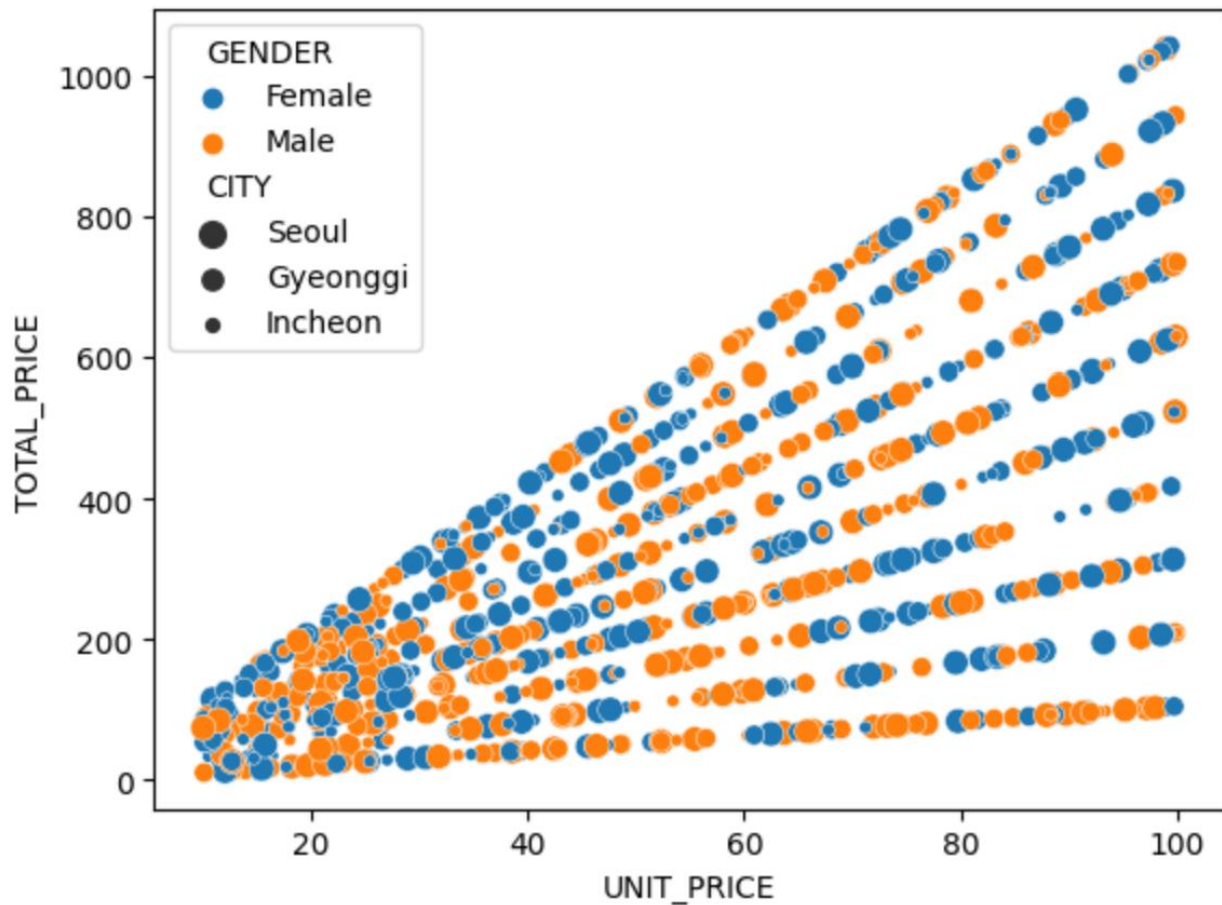
```
Out[29]: <Axes: xlabel='UNIT_PRICE', ylabel='TOTAL_PRICE'>
```



산점도

```
In [30]: 1 sns.scatterplot(data=df, x='UNIT_PRICE', y='TOTAL_PRICE', hue='GENDER', size='CITY')
```

```
Out[30]: <Axes: xlabel='UNIT_PRICE', ylabel='TOTAL_PRICE'>
```



AI/BigData 인텐시브 과정

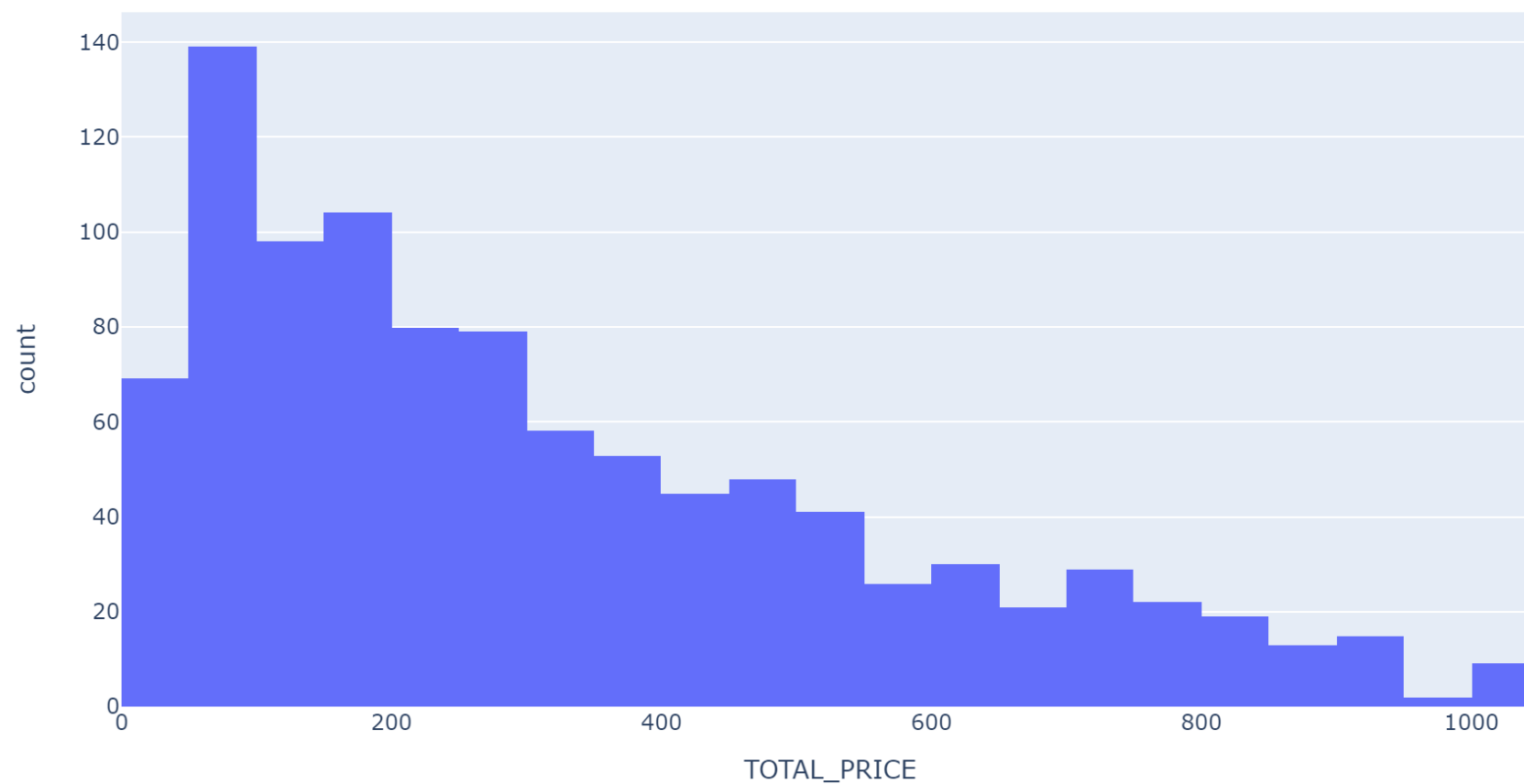
Section 1. 데이터 시각화

Section 1-3. plotly

히스토그램

```
In [31]: 1 import plotly.express as px
```

```
In [32]: 1 fig = px.histogram(data_frame=df, x='TOTAL_PRICE')  
2 fig.show()
```

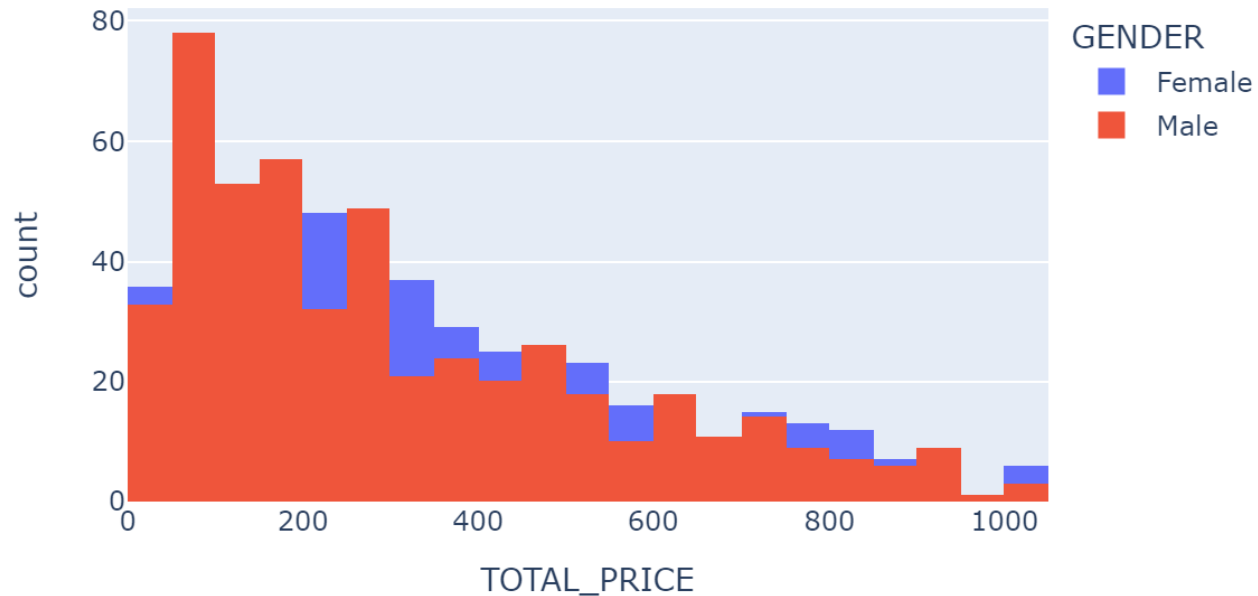


히스토그램

```
In [33]: ▶ 1 fig = px.histogram(data_frame=df, x='TOTAL_PRICE', color='GENDER', title='TOTAL PRICE GENDER')
          2 fig.update_layout(barmode='overlay',width=600,height=400)
          3 fig.show()
```

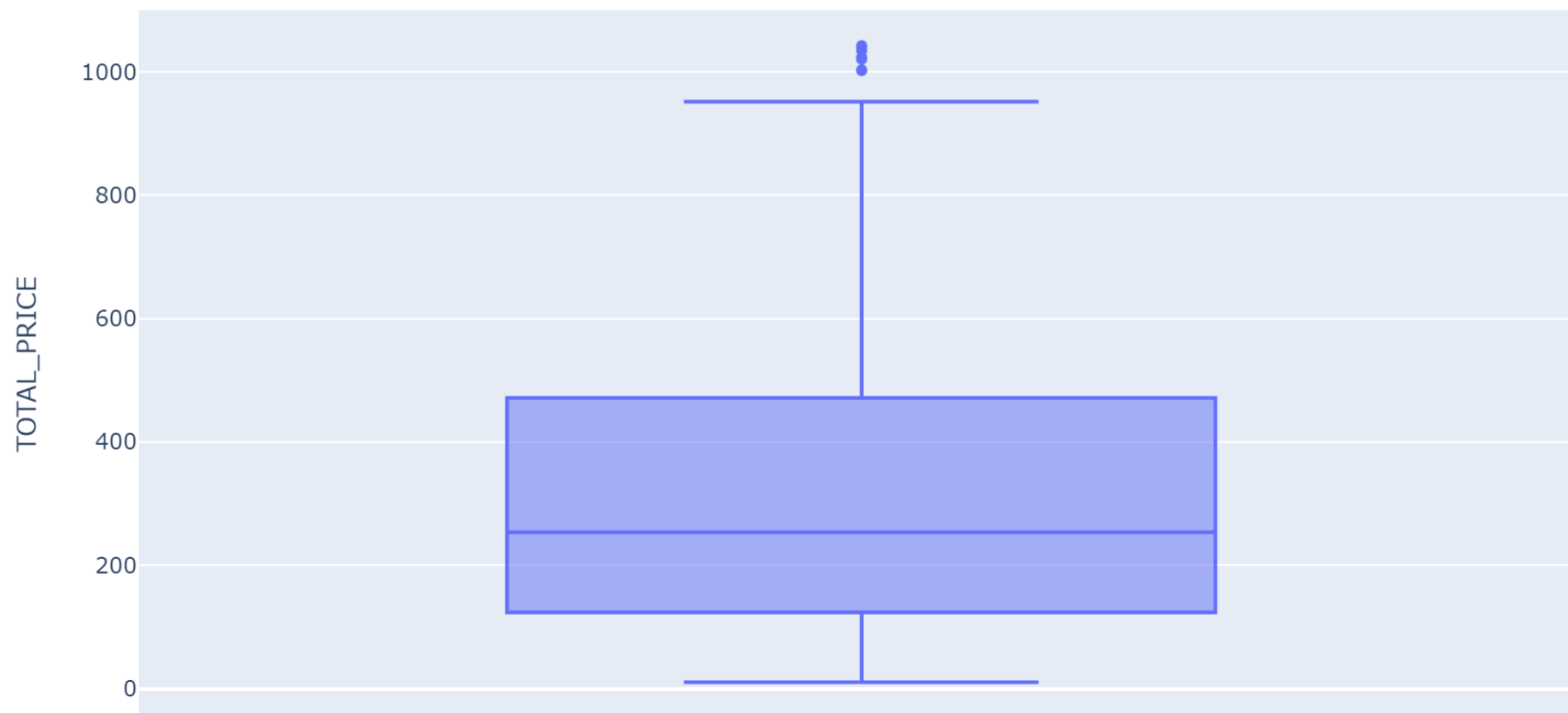
막대가 겹치게 fig 가로길이 fig 세로길이 타이틀

TOTAL PRICE GENDER



상자 그림(boxplot)

```
In [34]: ▶ 1 fig = px.box(data_frame=df, y='TOTAL_PRICE')  
2 fig.show()
```



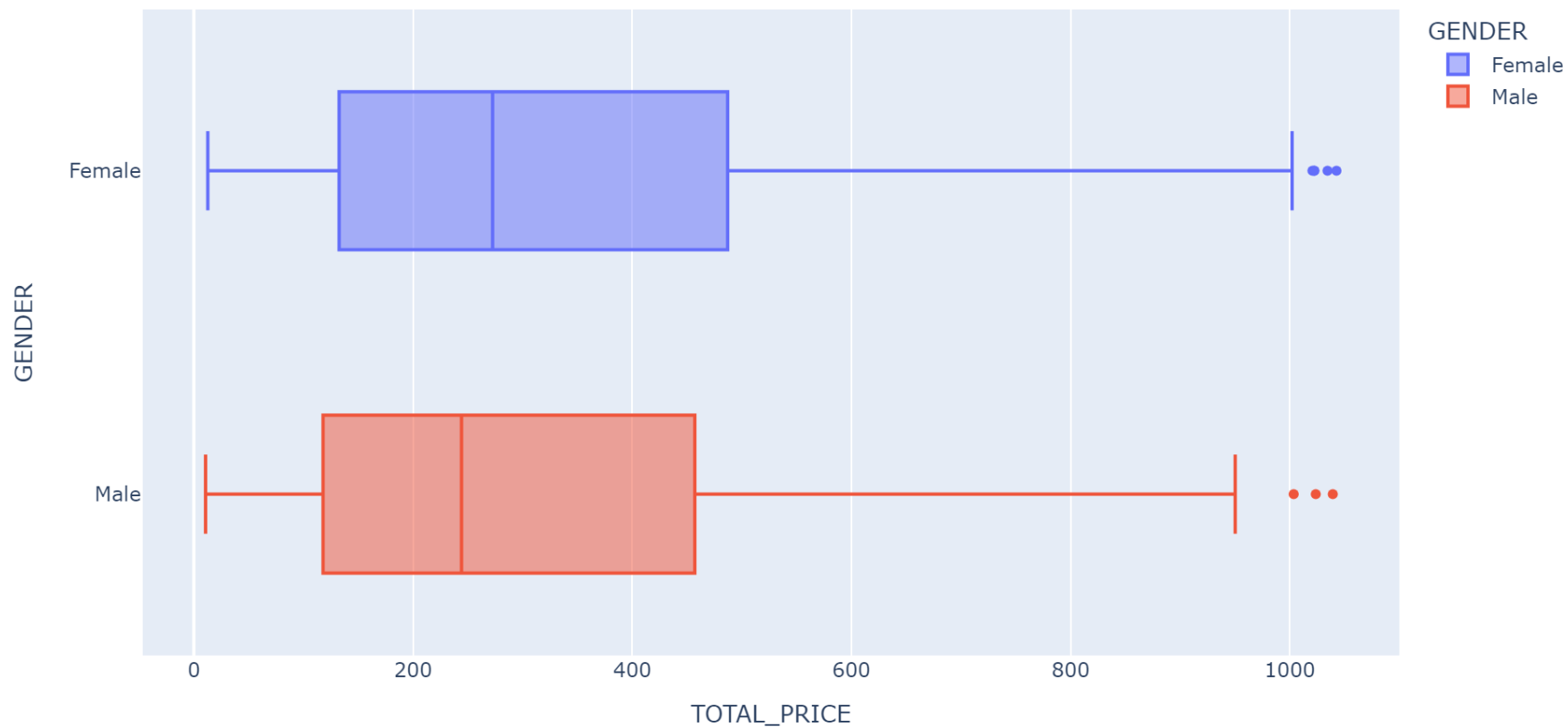
상자 그림(boxplot)

```
In [35]: ▶ 1 fig = px.box(data_frame=df, x='GENDER', y='TOTAL_PRICE', color='GENDER')  
2 fig.show()
```



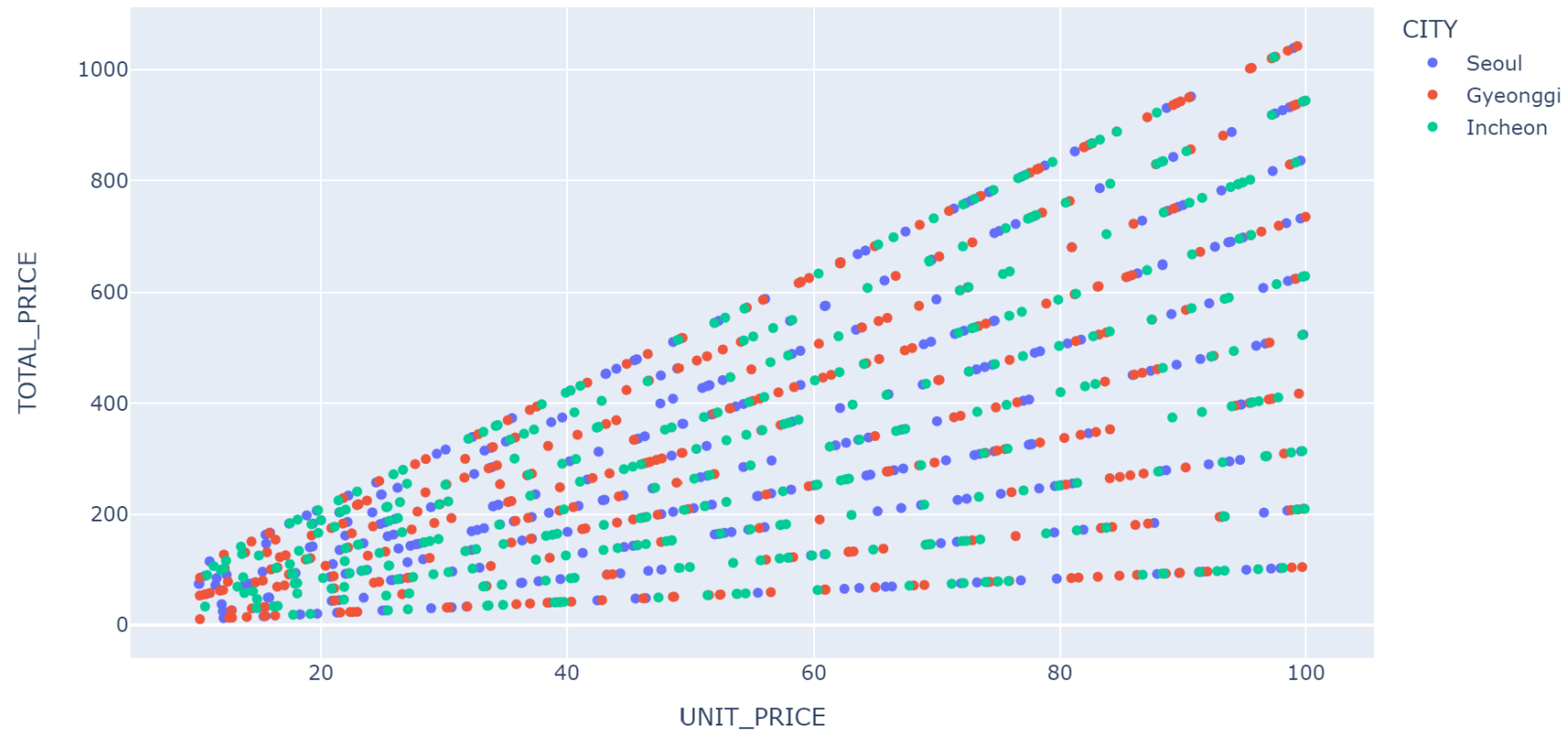
상자 그림(boxplot)

```
In [36]: 1 fig = px.box(data_frame=df, x='TOTAL_PRICE', y='GENDER', color='GENDER')
2 fig.show()
```



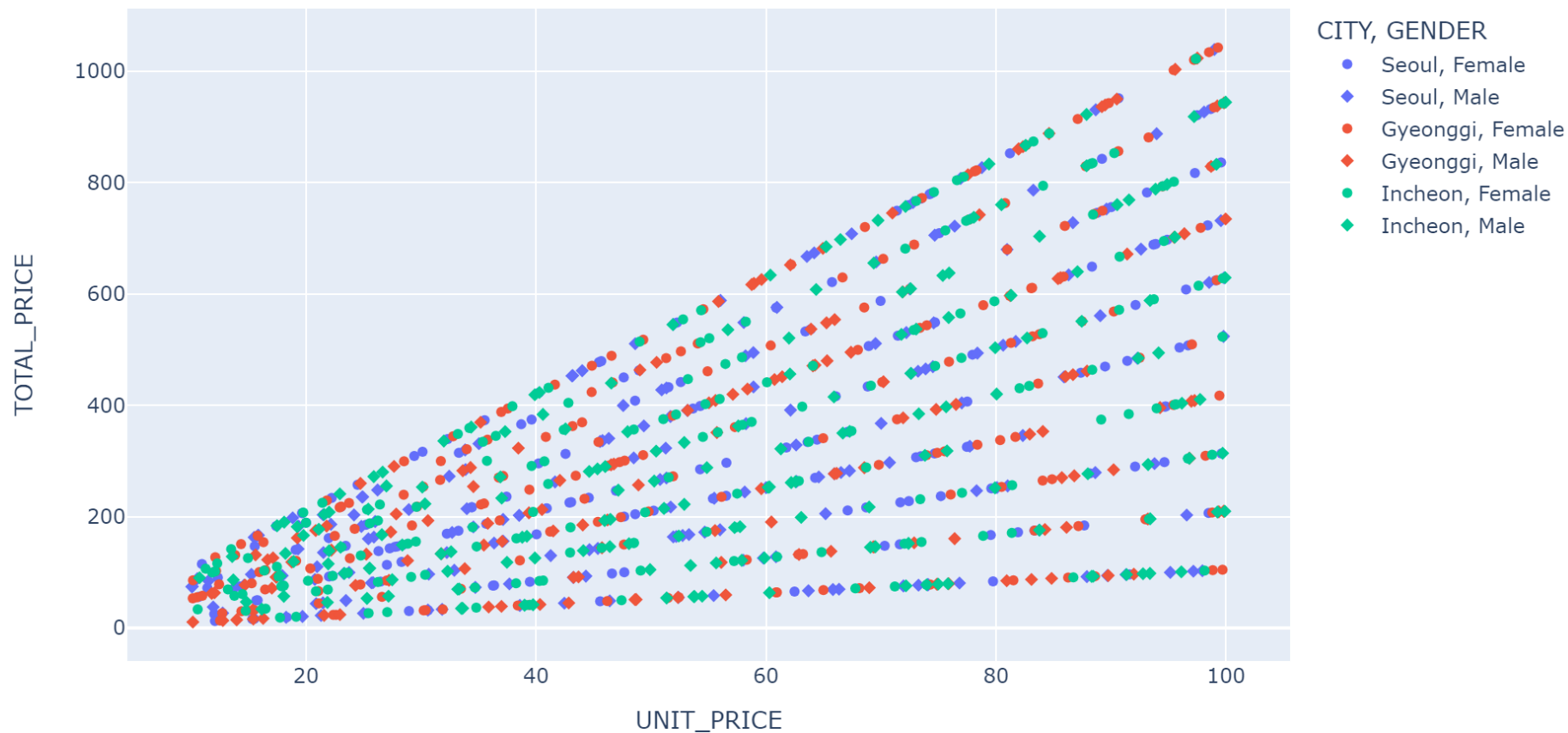
산점도(scatter plot)

```
In [37]: 1 fig = px.scatter(data_frame=df, x='UNIT_PRICE', y='TOTAL_PRICE', color='CITY')
2 fig.show()
```



산점도(scatter plot)

```
In [38]: 1 fig = px.scatter(data_frame=df, x='UNIT_PRICE', y='TOTAL_PRICE', color='CITY', symbol='GENDER')
2 fig.show()
```



파이 차트(pie chart)

In [39]: ▶

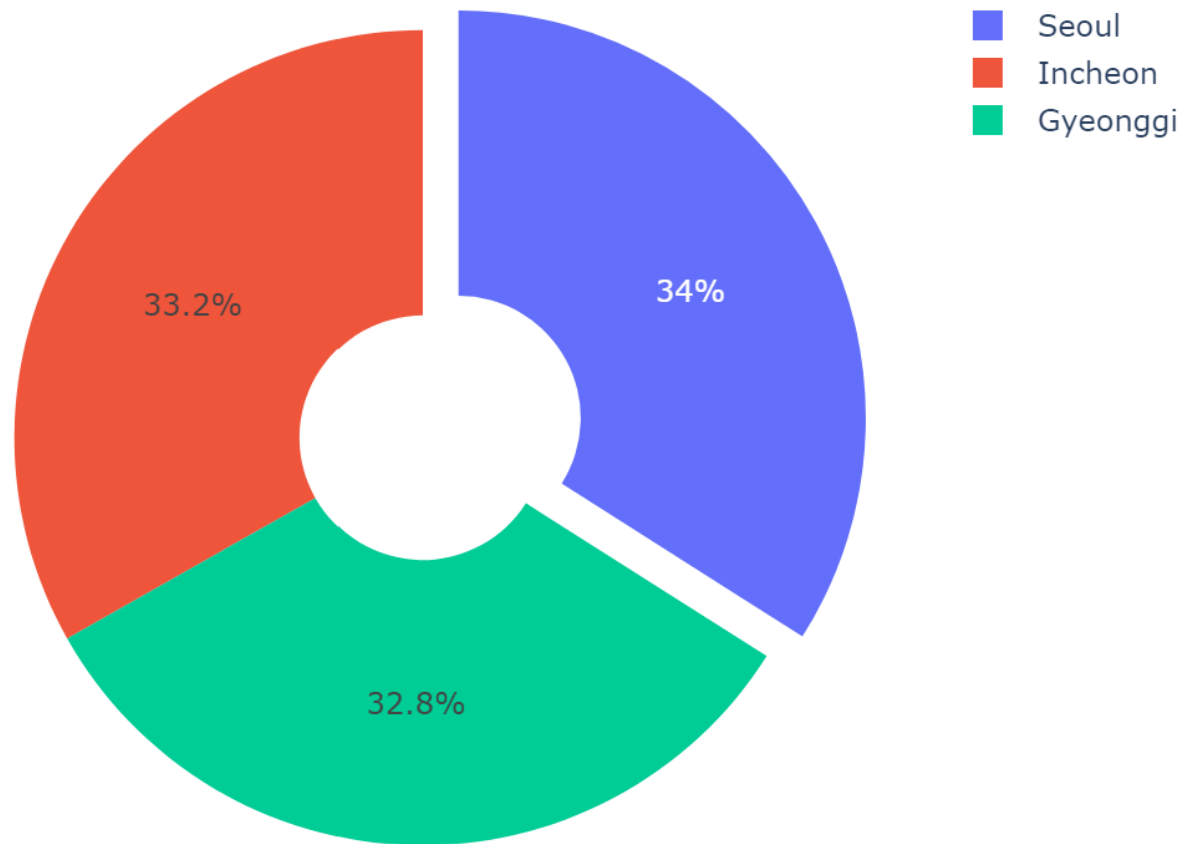
```
1 category_counts = df['CITY'].value_counts().reset_index()  
2 category_counts.columns = ['city', 'freq']  
3 print(category_counts)
```

	city	freq
0	Seoul	340
1	Incheon	332
2	Gyeonggi	328

파이 차트(pie chart)

In [40]: ▶

```
1 fig = px.pie(data_frame = category_counts, names='city', values='freq')  
2 fig.update_traces(hole=.3, pull=[0.1, 0, 0])  
3 fig.update_layout(width=600,height=500)  
4 fig.show()
```

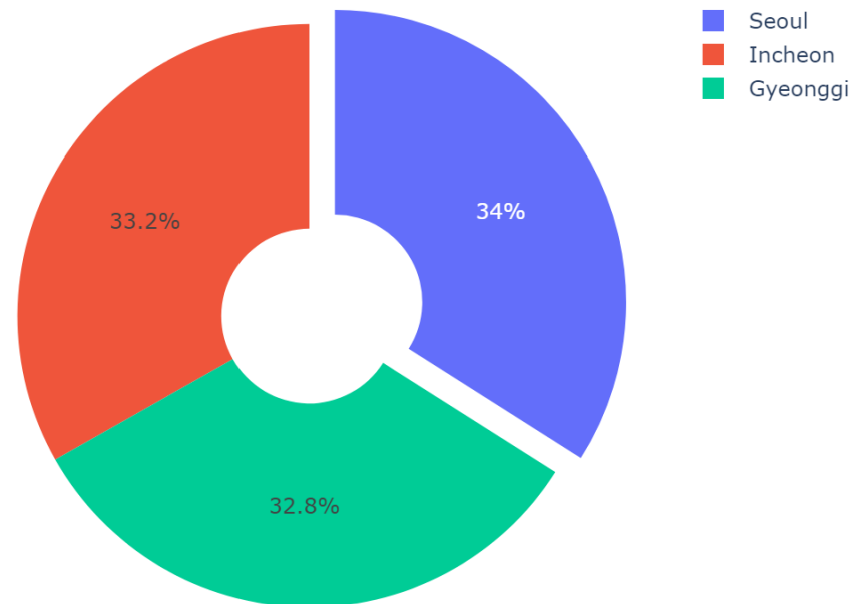


파이 차트(pie chart)

```
In [41]: ▶ 1 count_val = list(df['CITY'].value_counts())  
          2 print(count_val)  
  
[340, 332, 328]
```

```
In [42]: ▶ 1 count_idx = list(df['CITY'].value_counts().index)  
          2 print(count_idx)  
  
['Seoul', 'Incheon', 'Gyeonggi']
```

```
In [43]: ▶ 1 fig = px.pie(names=count_idx, values=count_val)  
          2 fig.update_traces(hole=.3, pull=[0.1, 0, 0])  
          3 fig.update_layout(width=600,height=500)  
          4 fig.show()
```



감사합니다.

Q & A